Outline

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- **Physical Design**
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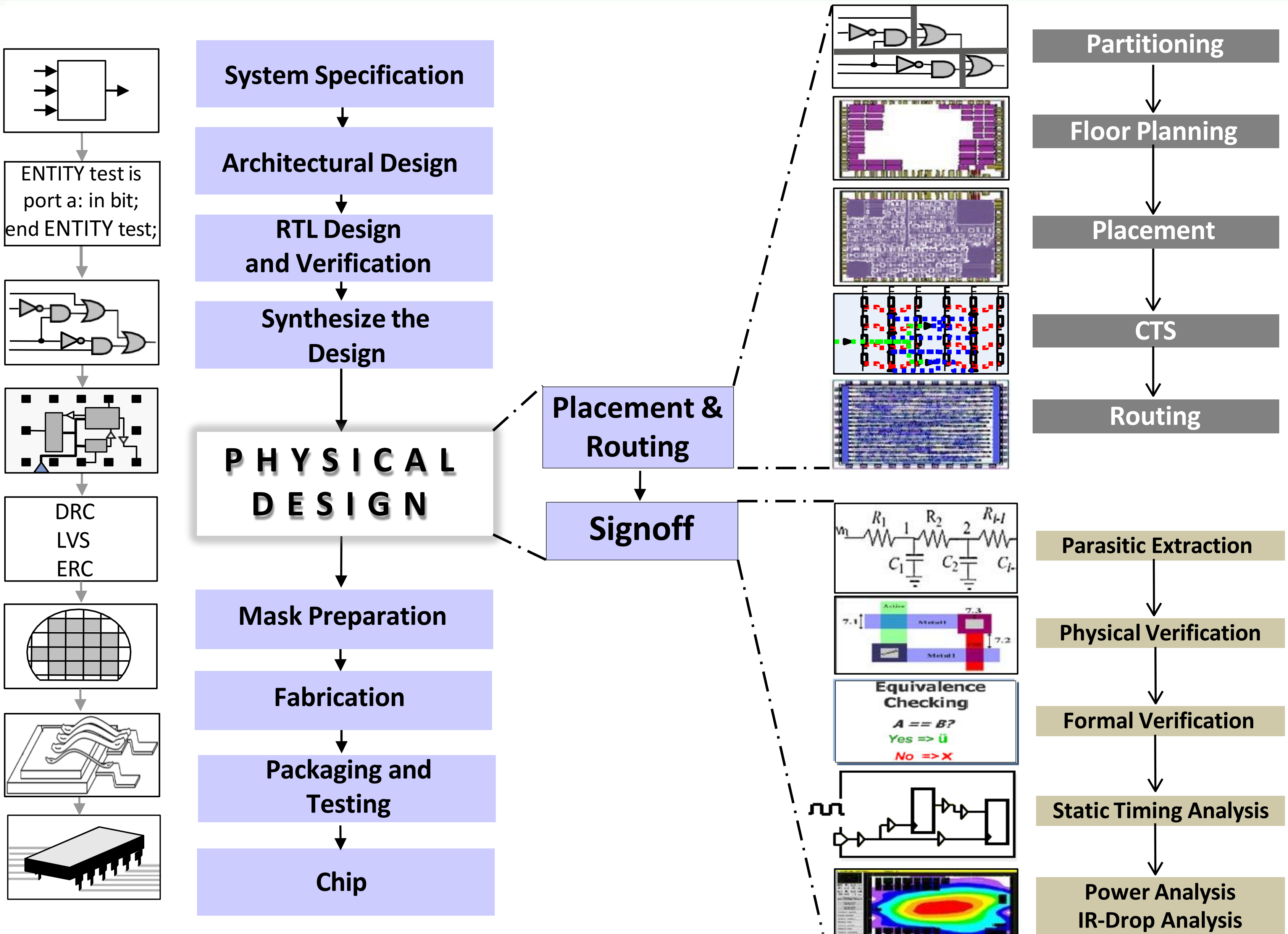
  - Physical Design Inputs
  - Physical Design Flow
    - Import Design & Partitioning
    - Floorplanning & Power planning
    - Placement & Placement Optimizations
    - CTS & CTS Optimizations
    - Routing & Routing Optimizations
    - Physical Verification (DRC, LVS, ERC)
    - DFM Checks
    - Formal Verification (LEC)
    - Parasitic Extraction (RC Extraction)
    - Timing Analysis (STA), Power Analysis & IR Drop Analysis
    - Tapeout

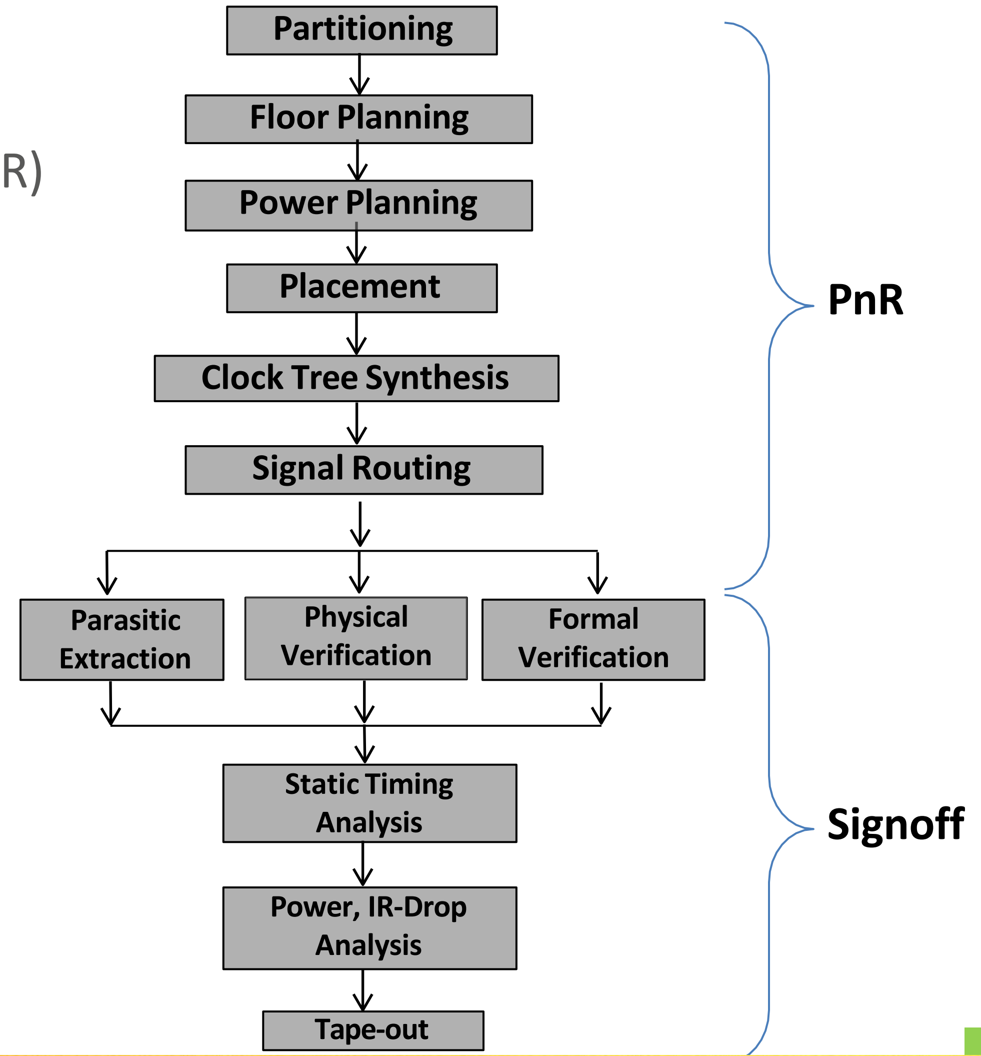
ASIC Design Flow



Physical Design Introduction

- Structural Representation to Physical Implementation

  i.e., Netlist to GDSII
- Stages
  - Placement and Routing (PnR)
  - Signoff
- Objectives
  - Timing
  - Congestion
  - Area
  - Power
- Possible Issues
  - Timing Violations
  - Congestion Issues
  - Design Rule Violations



Physical Design Inputs

In Physical Design mainly Six inputs are present

1. Logical libraries	--> format is .lib	--> given by Vendors
2. physical libraries	--> format is .lef	--> given by vendors
3. Technology file	--> format is .tf	--> given by fabrication peoples
4. TLU+ file	--> format is .TLUP	--> given by fabrication people
5. Netlist	--> format is .v	--> given by Synthesis People
6. Synthesis Design Constraints	-->format is .SDC	--> given by Synthesis People

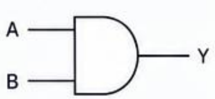
Physical Design Inputs

1. LIBERTY TIMING FILE (.LIB OR .DB) – THE TIMING LIBRARY

Liberty (.lib or .db) files describe the timing, power and functional characteristics of standard cells, macros and I/O cells.

A. Cell Logical View / The Timing Library

Represents how a cell behaves logically and electrically.
Used by synthesis, static timing analysis (STA) and power analysis.

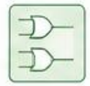


Example Liberty Snippet

```
cell (AND2_X1) {
  pin (A) { direction:input; }
  pin (B) { direction:input; }
  pin (Y) { direction: output;
   function: "A & B"; }
  timing () { ... }
}
```

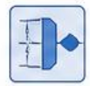
B. Types of Libraries

- **Standard Cell Library**
  Built from basic logic gates (AND, OR, NAND, FF, MUX, etc.)



- **Macro Library**
  Large pre-designed blocks (SRAM, PLL, DLL, ADC, etc.)

- **I/O Library**
  Pads, ESD cells, level shifters, bus interfaces, etc.



C. Gate Delay = f(input transition, output capacitance)

The delay of a cell depends on:

- Input transition time (slew)
- Output load capacitance (C_L)
- Operating conditions (V, T, PVT)

Example: Delay table for a cell pin (rise delay)

Input Transition (ns)	Output Capacitance (pF)				
	0.02	0.05	0.10	0.20	0.50
0.01	0.016	0.022	0.031	0.050	
0.05	0.020	0.028	0.039	0.063	
0.10	0.024	0.034	0.047	0.075	
0.20	0.022	0.042	0.058	0.092	

(Values shown are illustrative)

What's in a Liberty File?

- ✓ Cell & pin definitions
- ✓ Boolean function
- ✓ Timing arcs (rise/fall delays)
- ✓ Delay tables vs. input slew and output load
- ✓ Constraints (max cap, max tran, max fanout, etc.)
- ✓ Power (internal, switching, leakage power)
- ✓ Operating conditions, libs, PVT corners
- ✓ Bus definitions, groups, etc.

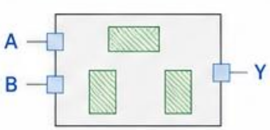
Used by tools: Synthesis, STA, Power Analysis, Formality

2. LIBRARY EXCHANGE FORMAT (LEF) – THE PHYSICAL LIBRARY

LEF (Library Exchange Format) files describe the physical and geometric information of library cells. Used by P&R tools for placement, routing and physical verification.

A. Cell Abstract View / The Physical Library

Abstract physical view of a cell.
Contains shape, size, pin location, obstruction, and placement/routing information.



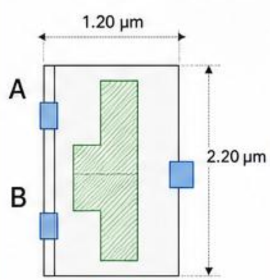
Example LEF Snippet

```
MACRO AND2_X1
 CLASS CORE ;
 ORIGIN 0 0 ;
 SIZE 1.20 BY 2.20 ;
 PIN A
  DIRECTION INPUT ;
  PORT
   LAYER M1 ;
   RECT 0.10 0.40 0.20 0.60;
  END
 END A
 ...
END AND2_X1
```

- ☑ Used for placement, routing, DRC, LVS, extraction
- ☑ Technology independent (abstract)
- ☑ Does not contain transistor-level details

B. Standard Cell LEF

Describes physical info of standard cells.
Small, placed in rows during P&R.



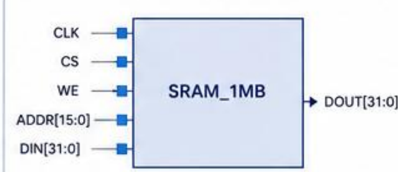
What it includes

- Cell size (width, height)
- Pin locations and shapes
- Metal layers information
- Obstructions
- Blockages
- Symmetry, site, class
- Spacing rules

Example Use: AND2_X1, INV_X2, DFF_X1, MUX2_X1, etc.

C. Macro LEF

Large hard macros (SRAM, PLL, ADC, SerDes, etc.).
Usually pre-placed or with placement restrictions.

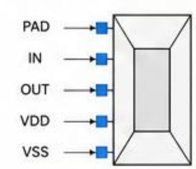


Includes:

- Macro size
- Pin locations
- Obstructions / blockages
- Keep-out margins
- Power/ground pins

D. IO LEF

Describes I/O cells and pads that connect the chip to the outside world.



Includes:

- Pad size and shape
- Pin locations
- ESD structures
- Ring/placement rules

Physical Design Inputs

- Technology Related files
  - Technology file
    - Technology file: format is .tf:
    - 1. It contains Name, Number conventions of layer and via
    - 2. It contains Physical, electrical characteristics of layer and via
    - 3. In Physical characteristics Min width, area, height are present.
    - 4. In Electrical characteristics Current Density is present.
    - 5. Units, Precisions, Colors and pattern of layer and via .
    - 6. Physical Design rules of layer and via
    - 7. In Physical Design rules Wire to Wire Spacing, Min Width between Layer and via are present..tech.lef (Cadence Format)
    - .tf - technology file (Synopsys Format)

  - Interconnect Parasitic file
    - Used for layer parasitic extraction
    - Contains Layer/ Via capacitance and resistance values in a Lookup Table (LUT) format
    - Also used to generate parasitic formats for the extraction tools (e.g. nxtgrd, captbl)
    - Extraction tool formats are more accurate than interconnect parasitic formats
    - .ict - Interconnect Technology Format (Cadence Format)
    - .itf - Interconnect Technology Format (Synopsys Format)
    - .ptf - Process Technology File (Mentor Graphics Format)

Physical Design Inputs

TLU+ files: format is .TLUP:

1. R,C parasitic of metal per unit length.
2. These(R,C parasitic) are used for calculating Net Delays.
3. If TLU+ files are not given then these are getting from .ITF file.
4. For Loading TLU+ files we have load three files .
5. Those are Max TLU+,MinTLU+,MAP file.
6. MAP file maps the .ITF file and .tf file of the layer and via names.

DC SDC :Format is .SDC :

These Constraints are timing Constraints and used for to meet timing requirements.

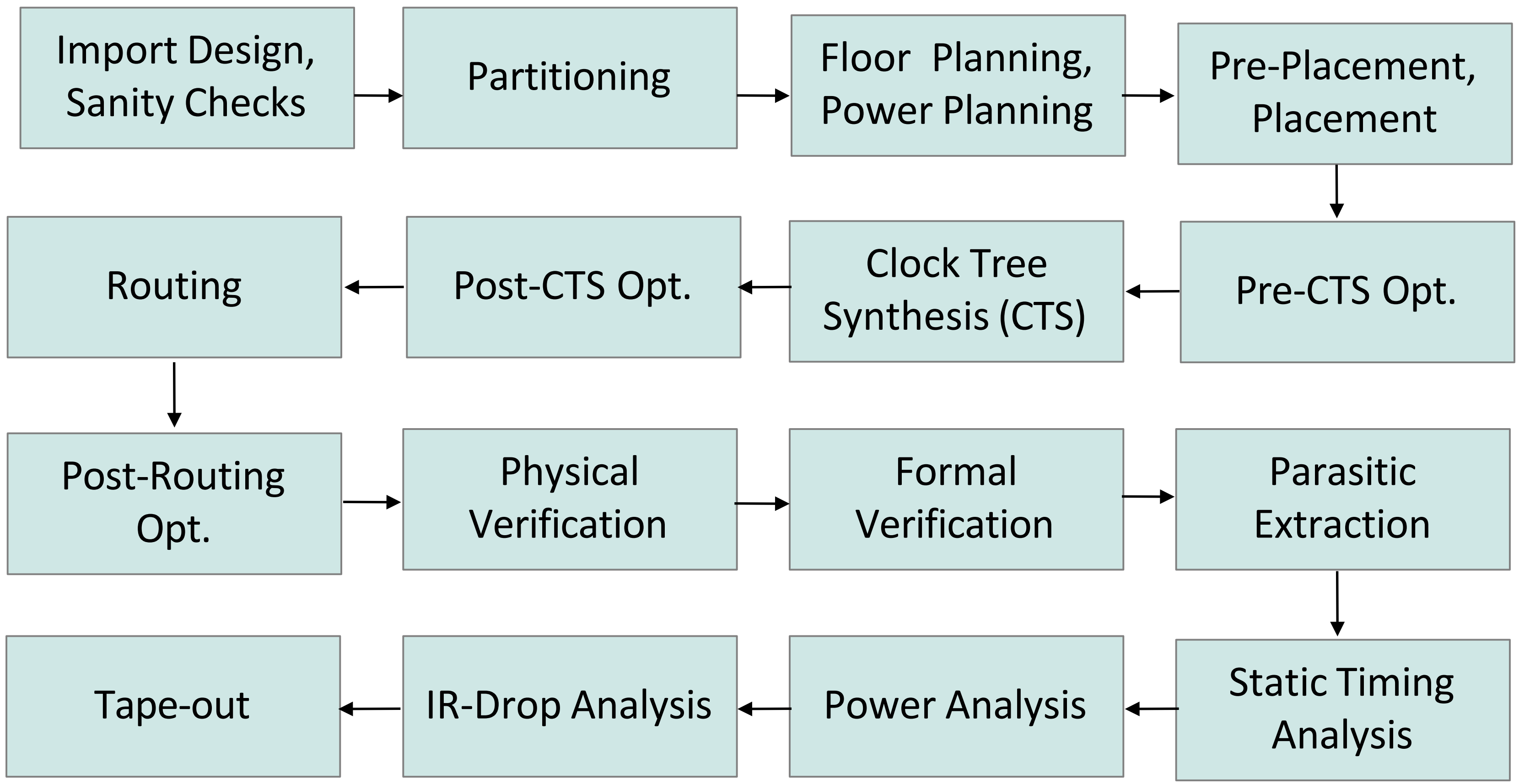
Constraints are

1. CLOCK DEFINITIONS: Create Clock Period. 2. Generated Clock Definitions
3. Input Delay 4. Output Delay 5. I/O delay 6. Max delay
7. Min Delay
8.--------------->Exceptions<------------------------ 9. Multi cycle path 10. False path
11. Half cycle path 12. Disable timing arcs
13. Case Analysis Multi cycle path, False path are Exceptions.

Physical Design Inputs

- Power Specification File
  - Power Modes & Power Domains
  - Tie Up supply & Tie Low supply
  - Power Nets & GND Nets
- Optimization Directives
  - Don't use
    - Cells that are not supposed to optimize
  - Size only/ use only
    - Upsizing/ Downsizing only with this list of cells
- Design Exchange Formats
  - List & locations of Components, Vias, Pins, Nets, Special nets
  - Die dimensions, Row definitions, Placement and Bounding Box Data, Routing Grids, Power Grids, Pre-routes
  - .def, .fp are the common formats
- Clock Tree Constraints/ Specification
  - Root Pin Definition
  - Insertion Delay (ID) and Skew Target
  - Maximum Capacitance/ Transition/ Fanout (DRVs)
  - Transition can be classified into Leaf Transition and Buffer Transition
  - No. of Buffer Levels (Tree depth)
  - List of Buffers/ Inverters for CTS
  - List of Through pin, Preserved Pin, Exclude Pin
  - NDRs can be defined in CTS Spec. for the Clock Tree Routing
  - Macro Models
- IO Information File
  - Pin/ Pad locations
  - Edge and order for IO Placement
  - .tdf, .io are common formats

Physical Design Flow



Import Design

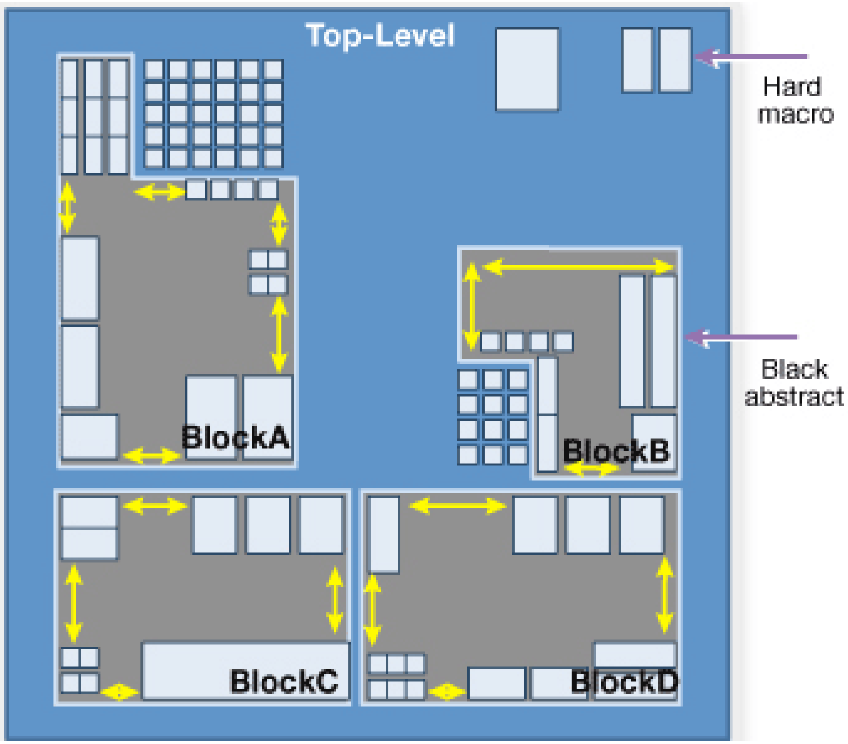
- Import Design
  - The following input files information are loaded to the PnR tool
    - Netlist (.v/ .vhd/ .edif)
    - Physical Libraries (.lef)
    - Timing Libraries (.lib)
    - Technology Files
    - Constraints (.sdc)
    - IO Info. File (optional)
    - Power Spec. File (optional)
    - Optimization Directives (optional)
    - Clock Tree Spec. File (optional at floorplan stage)
    - DEF/ FP (optional if floorplan is not done)
  - Core area is approximately calculated by the tool from the Netlist
  - While Importing, first we have to load the LEF files and then LIB files

Import Design

- Sanity Checks
  - Sanity Checks mainly checks the quality of netlist in terms of timing
  - It also consists of checking the issues related to Library files, Timing Constraints, IOs and Optimization Directives
  - Some of the Netlist Sanity Checks:
    - Floating Pins
    - Unconstrained Pins
    - Un-driven i/p Ports
    - Unloaded o/p Ports
    - Pin direction mismatches
    - Multiple drivers etc.
  - Other possible issues include Unconnected/ Wrongly Connected Tie-high/ Tie-low Pins and Power Pins (since Tie-up or Tie-down connectivity always through Tie-Cells)

Partitioning

- The Hierarchical Partitioning is done prior to Floorplan
- Partition can be done based on
  - Design Hierarchy
  - Timing Criticality
  - Functionality
  - Clock Domain
  - Design Files
  - Block Size
- Partitioning Inputs and Outputs by Registers
- Minimize Cross-Partition-Boundary IO



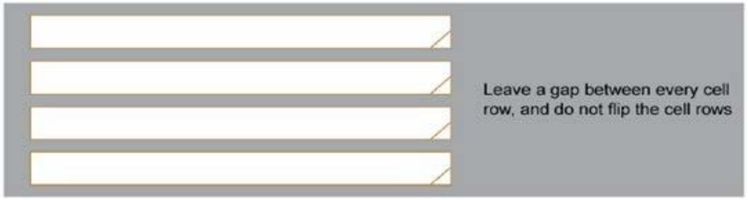
- For Sub-block designs, the Partitioning is not required
- For Full Chip only we need to design with Partitioning

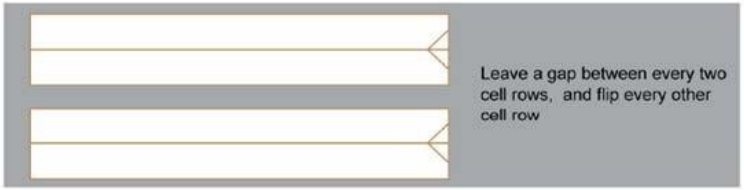
Floorplanning

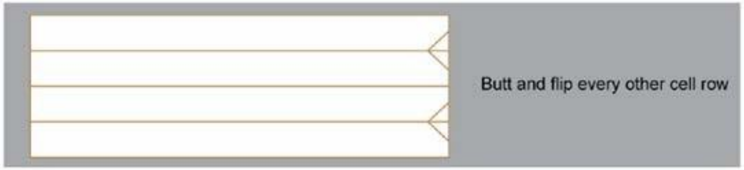
- Steps in Floorplan
  - Initialize with Chip & Core Aspect Ratio (AR)
  - Initialize with Core Utilization
  - Initialize Row Configuration & Cell Orientation
  - Provide the Core to Pad/ IO spacing (Core to IO clearance)
  - Pins/ Pads Placement
  - Macro Placement by Fly-line Analysis
  - Macro Placement requirements are also need to consider
  - Blockage Management (Placement/ Routing)

Floorplanning

- Initialization
  - Row Configuration
    - Slanting lines in the side of the cell rows denote the Cell Orientation

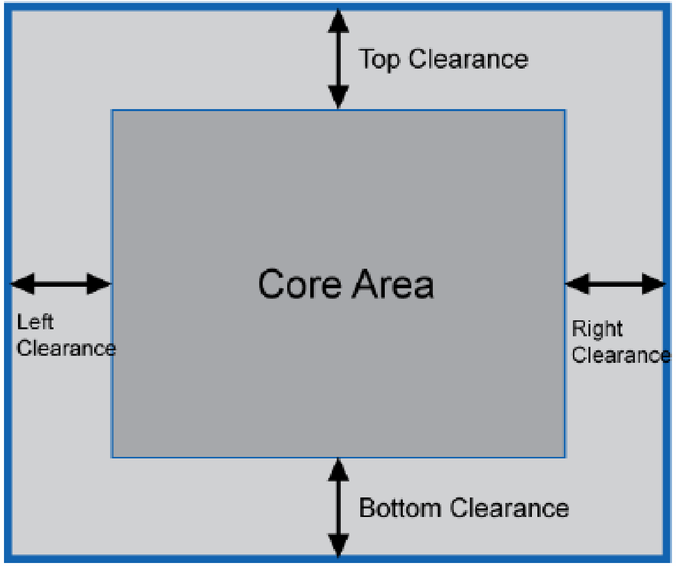






Most common because of better space utilization

  - Core to Pad/ IO spacing
    - Core to IO clearance
    - Used for Placing IOs and Power Ring



Floorplanning

Aspect Ratio

The Role of Aspect Ratio on the Design:

- The aspect ratio effects the routing resources available in the design
- The aspect ratio effects the congestion
- The floor planning need to be done depend on the aspect ratio
- The placement of the standard cells also effect due to aspect ratio
- The timing and there by the frequency of the chip also effects due to aspect ratio
- The clock tree build on the chip also effect due to aspect ratio
- The placement of the IO pads on the IO area also effects due to aspect ratio

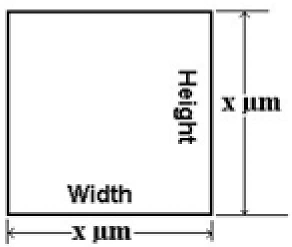


Fig.1: Aspect Ratio = 1.0

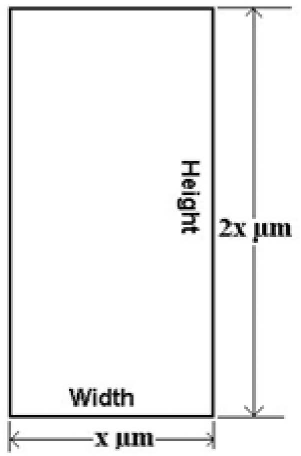


Fig.2: Aspect Ratio = 2.0

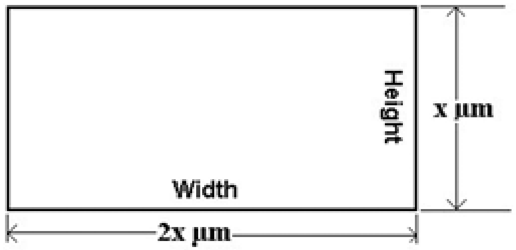
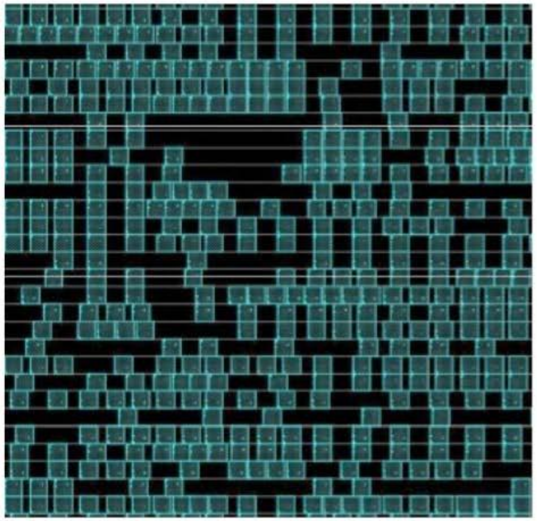


Fig.3: Aspect Ratio = 0.5

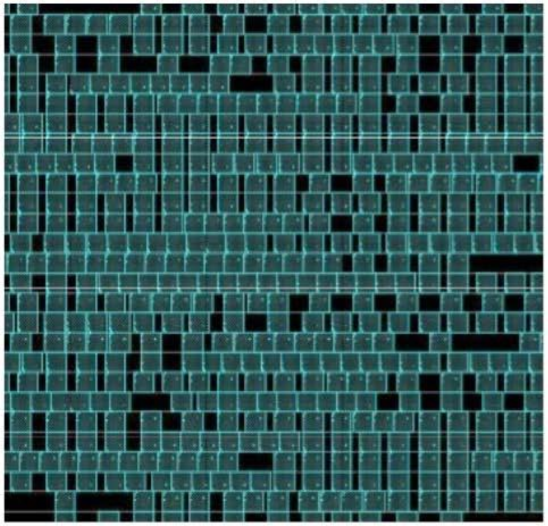
$$\text{Aspect Ratio} = \frac{\text{Height of the Core}}{\text{Width of the Core}}$$

Floorplanning

– $$\text{Utilization} = \frac{\text{Total Standard Cell Area} + \text{Macro Area}}{\text{Total Core Area}} \times 100\ \%$$



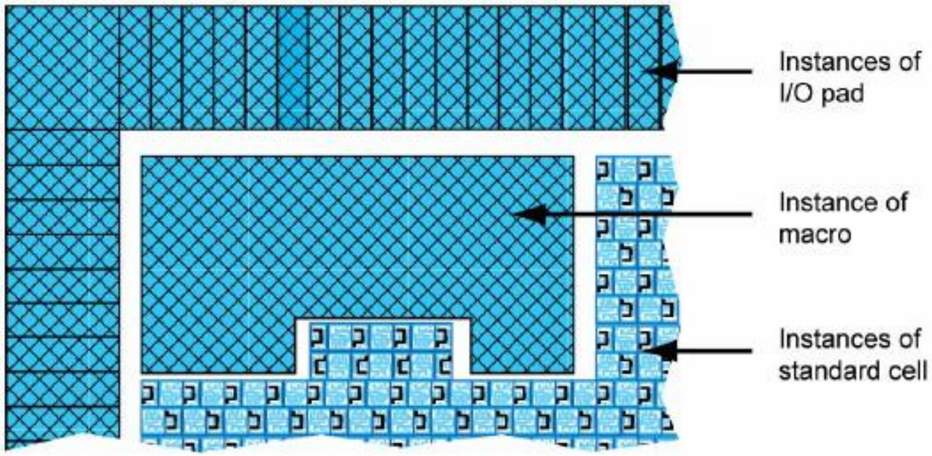
Low standard-cell utilization



High standard-cell utilization

Floorplanning

- IO Placement
  - Chip Level its IO Pads and Block Level its IO Pins
  - Pin is a logical entity and is a property of a Port
  - Port is a physical entity and a Port have only 1 Pin associated with it
  - Netlist will have Pins and Layout will have Ports
  - Unplaced Port is not represented in the Layout
  - Different types of IOs
    - Signal Pads/Pins
    - Core Power Pads/Pins
    - IO Power Pads/Pins
    - Corner Pads (Doesn't hold any logic, provides IO Pad Ring connectivity)
    - Filler Pads (Fill the gaps between IO pads to get the Ring Connectivity)
  - Physical-only pads that are not part of the input Gate level Netlist need to be inserted prior to reading IO constraints



Floorplanning

IO Placement

- IO Pads enables the design to operate at different voltages with the help of Level Shifters, Pre-Drivers (at Core Voltage) Post-Drivers (at IO Voltage)
- No of Core Power Pads needed:

$$\frac{\textbf{Total Dynamic Power}}{\textbf{No.of Sides x Core Voltage x Maximum allowable Current of IO Pad}}$$

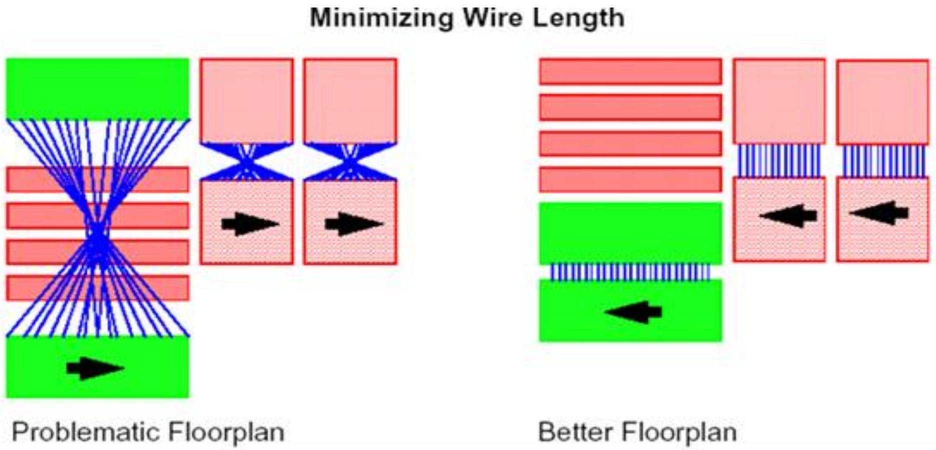
- There will be 1 Core GND Pad along with every Core Power Pad
- No. of IO Power Pads needed:

Thumb Rule: 1 pair of IO Power Pads for every 4 to 6 Signal Pads

Floorplanning

- Macro Placement
  - Fly-line Analysis (For Connectivity information)
  - Macro keep-out (For Uniform Standard Cell Region)
  - Channel Calculation (Critical for Congestion and Timing)
  - Avoid odd shaped area for Standard Cells
  - Funnel shaped Macro Placements are preferred
  - Fix the Macro locations, so that tool wont alter during Optimization
  - Spacing between Macro:

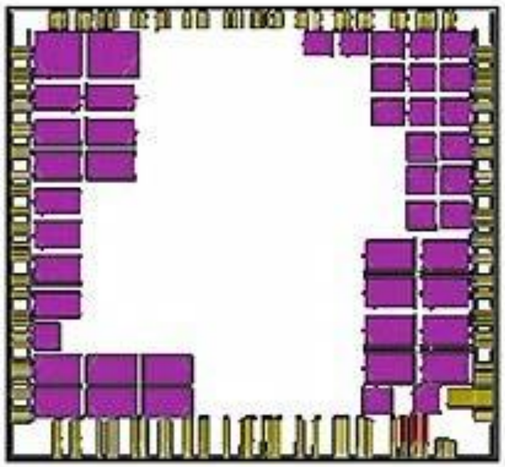
$$\frac{\text{Pitch of the Routing Layers x No.of pins to be Routed}}{\text{Available Routing Layers in the prefered direction}} + \text{Buffer Spacing}$$



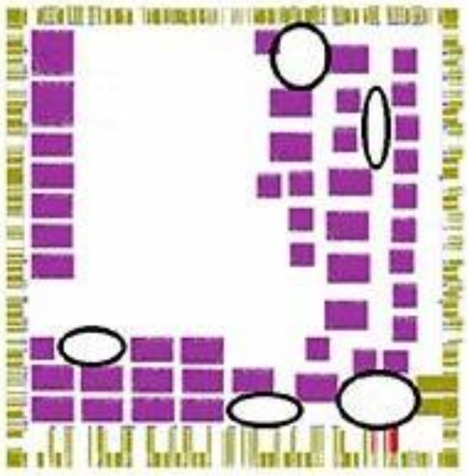
Floorplanning

- Macro Placement Tips
  - Place macros around chip periphery, so that core area will be clustered
  - Consider connections to fixed cells when placing Macros
  - In advanced Technology Nodes Macro Orientation is fixed since the Poly Orientation can’t vary, so there will be restrictions in Macro Orientation
  - Reserve enough room around Macros for IO Routing
  - Reduce open fields as much as possible
  - Provide necessary Blockages around the Macro

Homogeneous Standard Cell Area With Aligned Macros



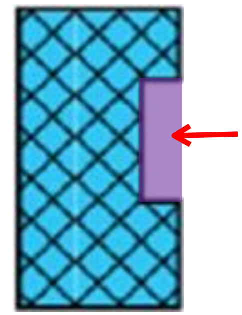
Irregular Macro Placement With Traps for Standard Cells



Floorplanning

- Blockages
  - Placement Blockage & Routing Blockage
  - Both of the Blockages can again be classified as-
    - Hard, Soft and Partial Blockages
  - Hard Blockage
    - Complete Standard Cell Blockage
  - Soft Blockage
    - Non-Buffering Blockage
  - Partial Blockage
    - Partial Standard Cell Blockage and is used to avoid congestion
    - We can Block Standard Cells as per the required percentage value
  - Keep-out/ Halo
    - Halo is similar to Soft Blockage (Terminology in Cadence EDI)
    - Its basically a keep-out Macro margin
    - Halo respects Macro while other Blockages respect location i.e., even if Macro is moved Halo also moves along with it

Rectilinear Macro Without Blockage

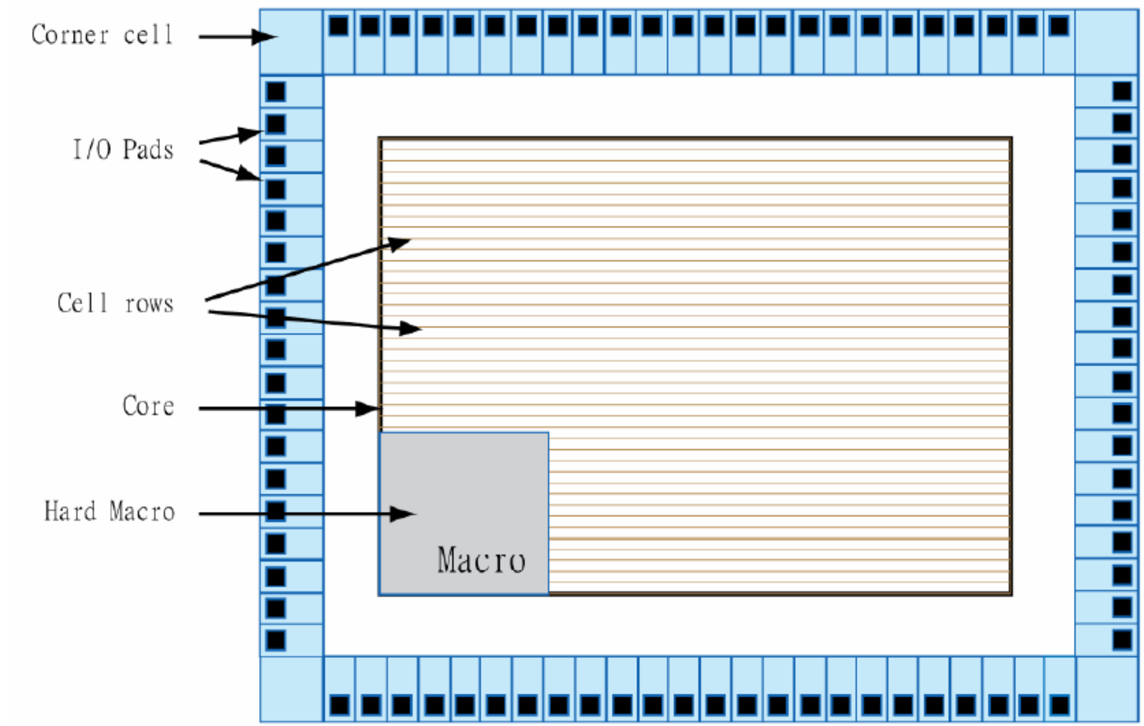


With Blockage

Halo around Macro

Floorplanning

- Issues arises due to bad Floorplan
  - Congestion near Macro Pins/ Corners due to insufficient Placement Blockage
  - Std. Cell placement in narrow channels led to Congestion
  - Macros of same partition which are placed far apart can cause Timing Violation



Floorplan done with 1 Macro

Power planning

- Power Plan
  - To connect Power to the Chip by considering issues like EM and IR Drop
  - Power Routing also called Pre-Routing
  - Pre-Routing includes creating Power Ring, Stripes/Mesh/Grid, and Standard Cell Power Rails
  - Power Planning also includes Power Via insertion
  - IO Rings are established through IO Cell abutment and through IO Filler Cells
  - Power Trunks are constructed between Core Power Ring and Power Pads
  - Trunk is a piece of metal that connects IO Pad and Core Ring
  - Technical information required for Power Planning:
    - Total Dynamic Power info. will get from Compiler
    - Technology File will provide Current Density (J_{MAX})
    - LEF will prove the Metal Layer width
    - Technology Library will provide Core Voltage

Power planning

- Levels of Power Distribution
  - Rings
    - V_{DD} and V_{SS} Rings are formed around the Core and Macro
  - Stripes
    - Carries V_{DD} and V_{SS} around the chip
    - Carries V_{DD} and V_{SS} from Rings across the chip
    - Power Stripes are created in the Core Area to tap power from Core Rings to the core area
  - Rails (Special Route)
    - Connect V_{DD} and V_{SS} to the standard cell
    - Standard Cell Rails are created to tap power from Power Stripes to Std. Cell Power/Ground Pins
  - Power Vias
    - Insert all Power Vias between Ring & Grid, Grid & Rail and Vertical Grid & Horizontal Grid
  - Trunks
    - Connects Ring to Power Pad

Power planning

- Power Plan: Calculations
  - Total Dynamic Core Current =

$$\frac{\textbf{Total Dynamic Core Power}}{\textbf{Core Voltage}}$$

  - Pad to Core Trunk Width =

$$\frac{\textbf{Total Dynamic Core Current}}{\textbf{No. of sides x JMAX x Current Sources per side}}$$

  - Core Ring Width =

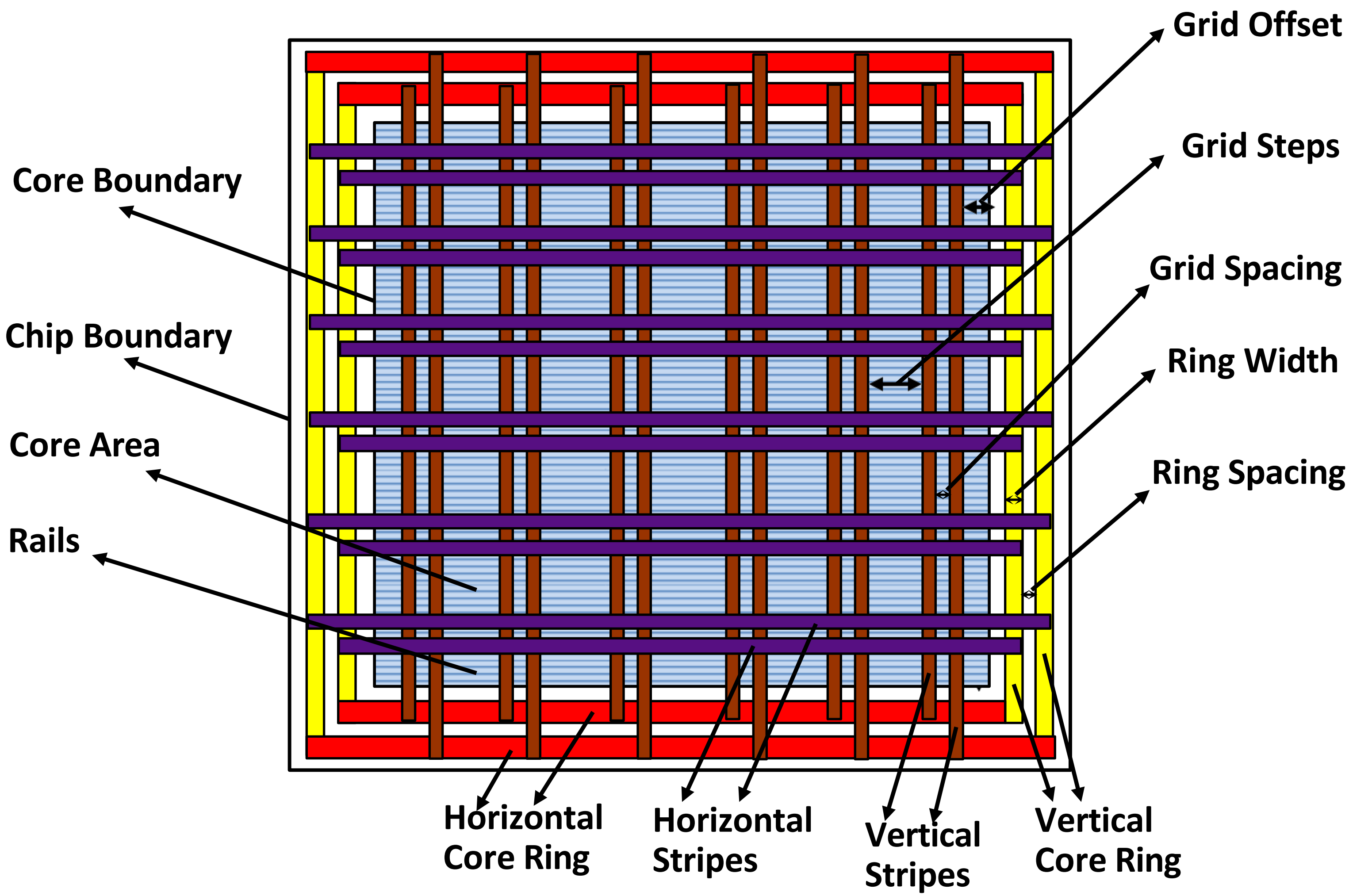
$$\frac{\textbf{Total Dynamic Core Current}}{\textbf{2 x No. of sides x JMAX of the Metal Layer for Ring x Current Sources per side}}$$

  - Power Stripes Spacing =

$$\frac{\textbf{Core Width} - (\textbf{No. of Stripes x Width of the Stripe})}{\textbf{No. of Stripes} + 1}$$

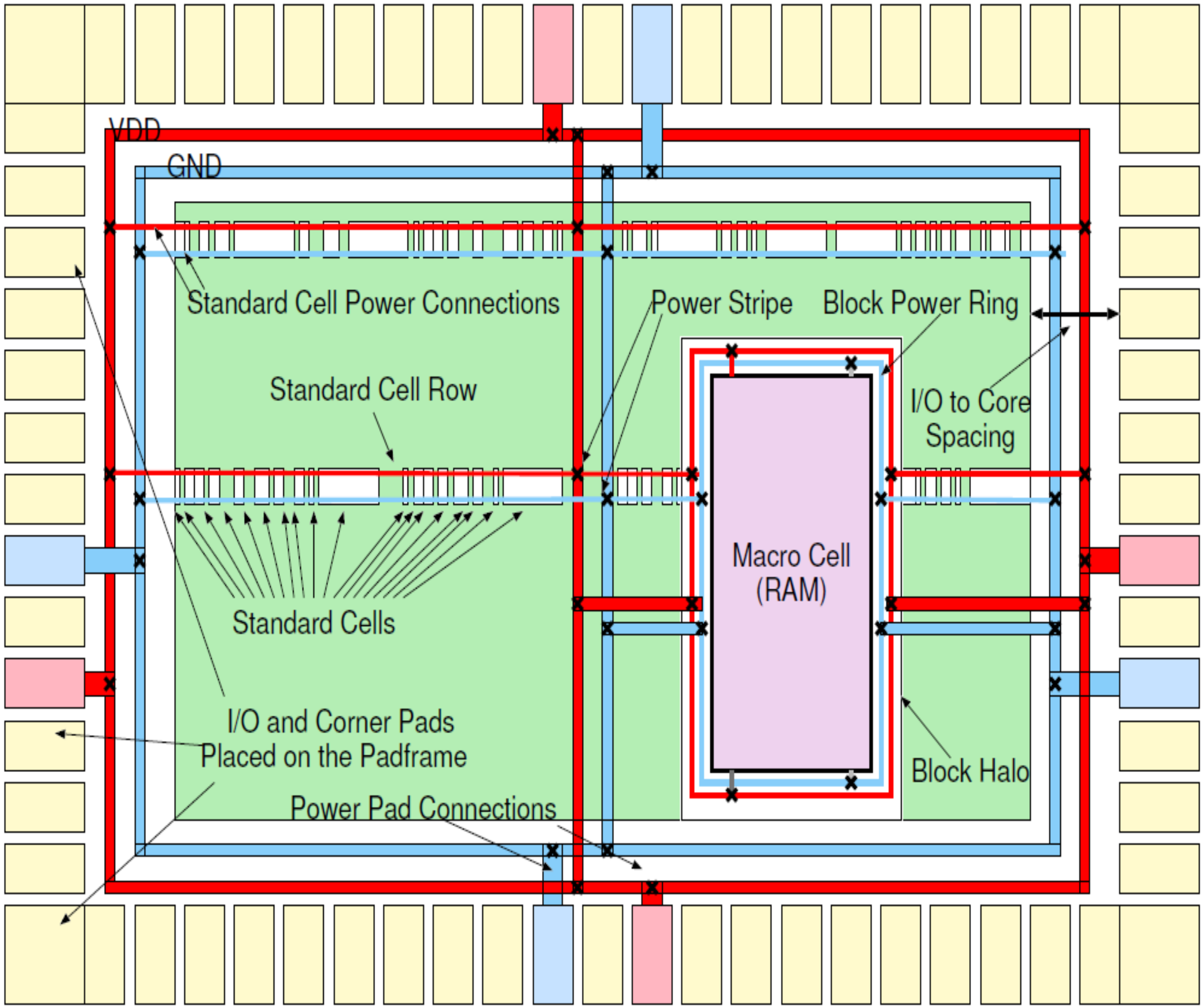
Power planning

- Sub-block Configuration



Power planning

- Full Chip Configuration



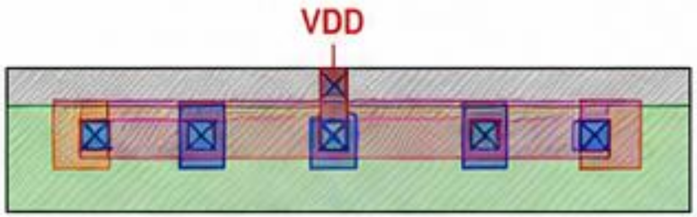
- Save Floorplan (.def / .fp)

Pre-Placement

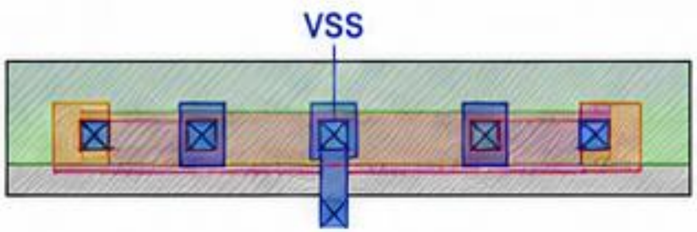
PHYSICAL-ONLY CELLS (WELL TAPS, END CAPS)

WELL TAP CELLS

- Connect wells and substrate to power rails
- Prevent latch-up by tying wells to VDD/VSS
- No signal connectivity



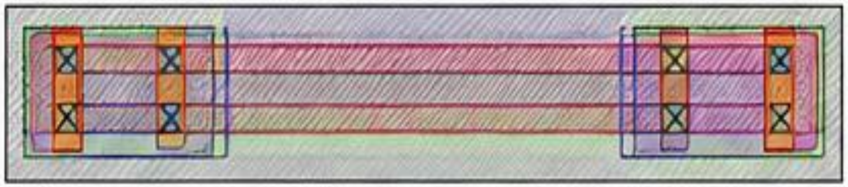
P+ WELL TAP (N-WELL TO VDD)



P+ SUBSTRATE TAP (SUBSTRATE TO VSS)

END CAP CELLS

- Placed at the end of rows to avoid gaps
- Prevent DRC violations in Well & Implant layers
- Ensure well continuity and proper tie-off



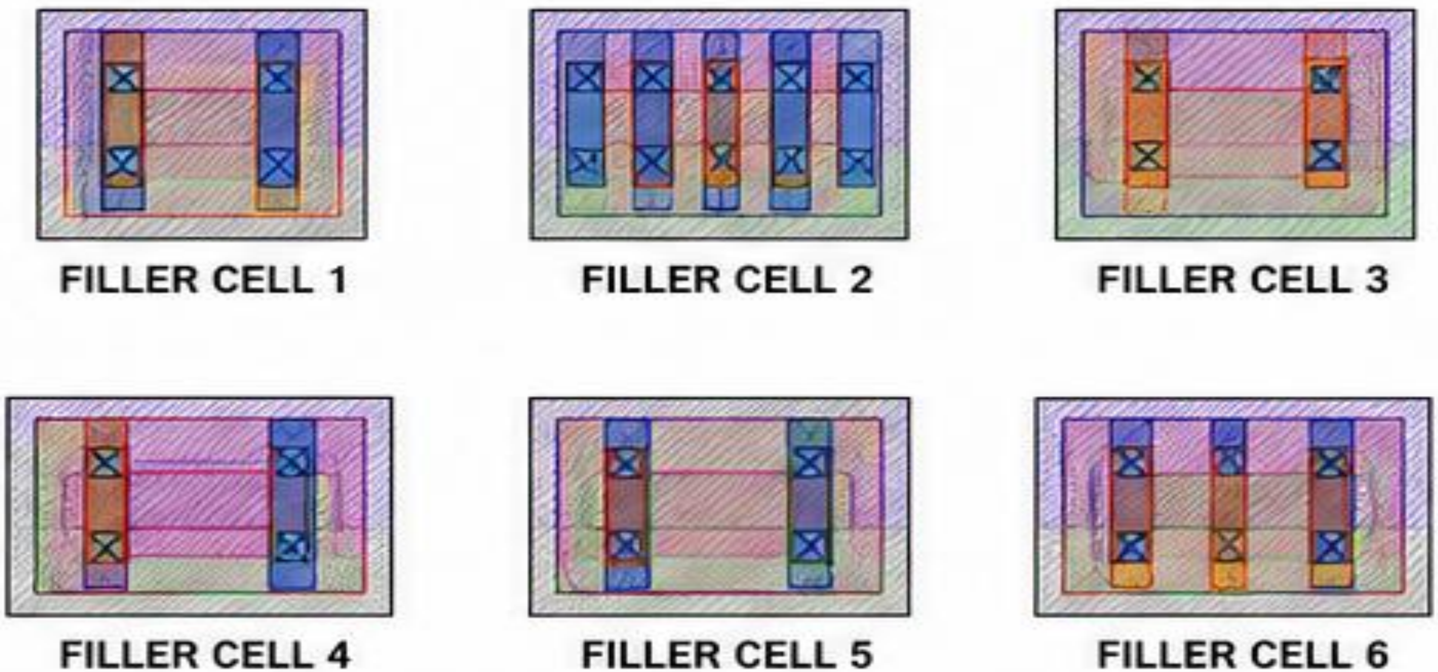
END CAP (N-WELL)

END CAP (P-WELL)

Pre-Placement

FILLER CELLS

- Used to fill whitespace between standard cells
- Improve routability and density
- No signal connectivity



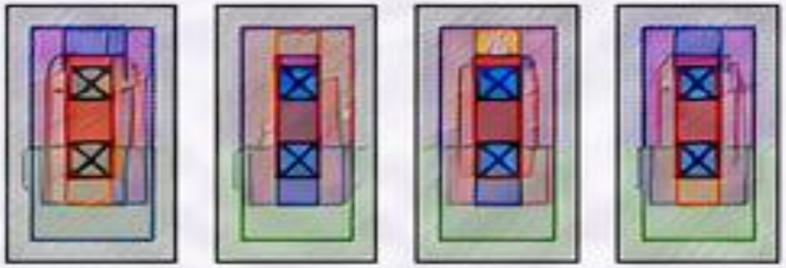
SPECIAL CELLS (SPARE CELLS, DECAP CELLS)

SPARE CELLS

- Used for ECO (Engineering Change Orders)
- Replaces or adds logic without re-spinning the chip
- No signal connectivity

SPARE CELL INFORMATION

Cell Name	: SPARE_X1
Cell Type	: Physical-Only (Spare)
Function	: Placeholder for future use
Connectivity	: Power (VDD, VSS) only
Pins	: VDD, VSS
Drive Strength	: N/A
Usage	: ECO / Scan chain repair / Logic fix
Placement	: Evenly distributed across the core
Size	: Same as standard cell site
Signal Pins	: None



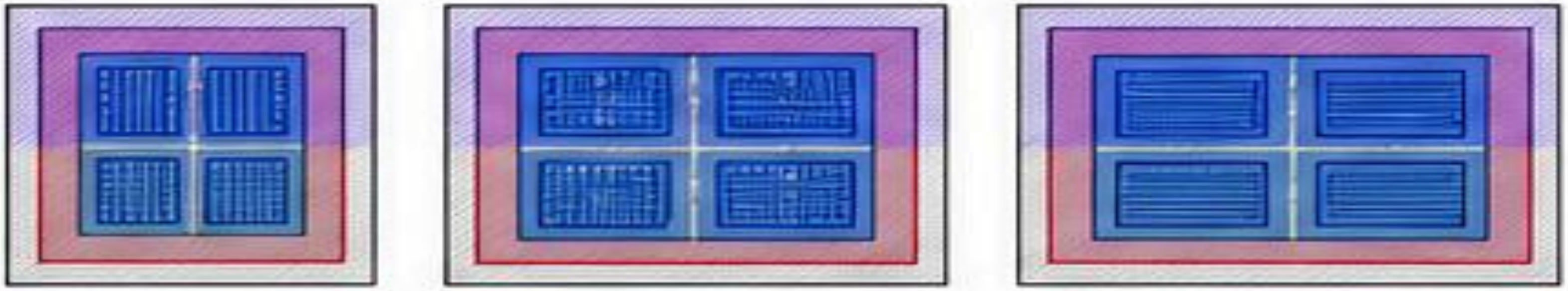
EXAMPLES OF SPARE CELLS

Spare cells are pre-placed during initial design to save area and routing effort during ECO. They have no functional logic.

SPECIAL CELLS (SPARE CELLS, DECAP CELLS)

DECAP CELLS

- Decoupling capacitors reduce Instantaneous Voltage Drop (IVD)
- Stabilize power supply for large drivers and switching activity
- Place closer to Power Pads or large Drivers



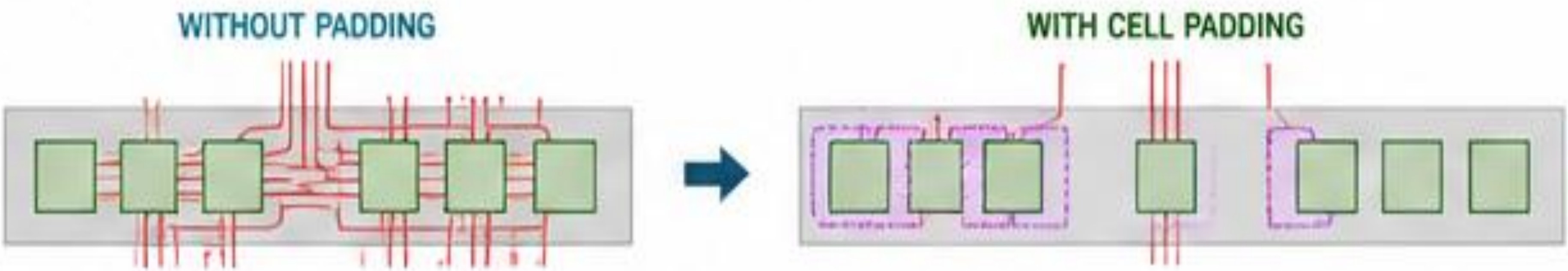
EXAMPLES OF DECAP CELLS

DECAP CELL INFORMATION

Cell Name	:	DECAP_XxY
Cell Type	:	Decoupling Capacitor
Function	:	Reduce IR drop / Noise
Connectivity	:	Power (VDD, VSS) only
Pins	:	VDD, VSS
Capacitance	:	X fF (e.g., 10fF, 20fF, 30fF...)
Placement	:	Near power pads / large drivers

CELL PADDING

- Cell Padding is done to reserve space for avoiding Routing Congestion
- Cell Padding adds Hard Constraints to Placement
- The Constraints are honored by Cell, Legalization, CTS, and Timing Optimization

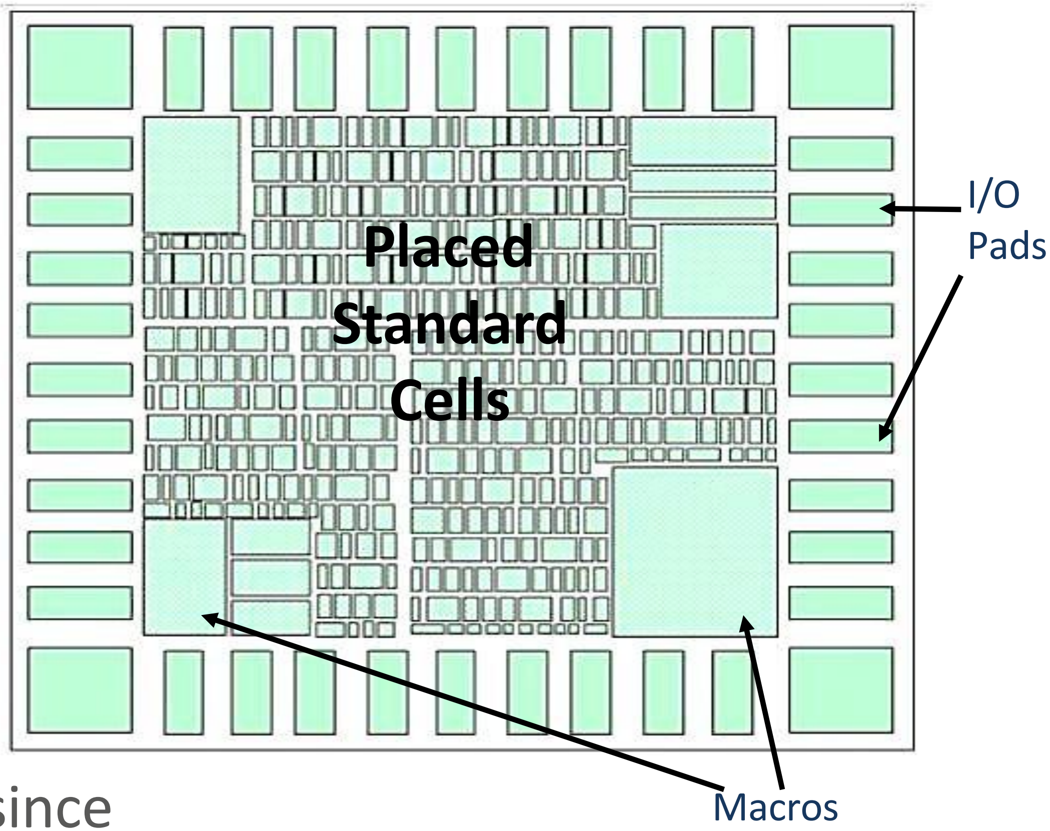


SUMMARY

- **Well Tap Cell:** Like grounding wires in a building to prevent electrical shocks.
- **End Cap Cell:** Like closing the boundary edges to avoid gaps in construction.
- **Filler Cell:** Like small fillers in a puzzle to remove empty spaces.
- **Spare Cell:** Like spare parts kept for future repairs.
- **Decap Cell:** Like extra batteries to supply instant power during high demand.
- **Cell Padding:** Like keeping buffer space around crowded areas to avoid congestion.

Placement

- Automated Standard Cell Placement for placing the Standard Cells in Placement Tracks
- Placement Objectives
  - Total wire length
  - Routability
  - Performance
  - Power
  - Heat distribution
- Timing checks only with slow corners at Placement stage
- Only Setup Time check, since buffers are getting added during Clock Tree Synthesis



Global Placement

Initial Placement / Global Placement / Course Placement

- During the coarse placement, the tool determines an approximate location for each cell according to the timing, congestion and multi-voltage constraints.
- The placed cells don't fall on the placement grid and might overlap each other. Large cells like RAM and IP blocks act as placement blockages for standard cells.
- Coarse placement is fast and sufficiently accurate for initial timing and congestion analysis.

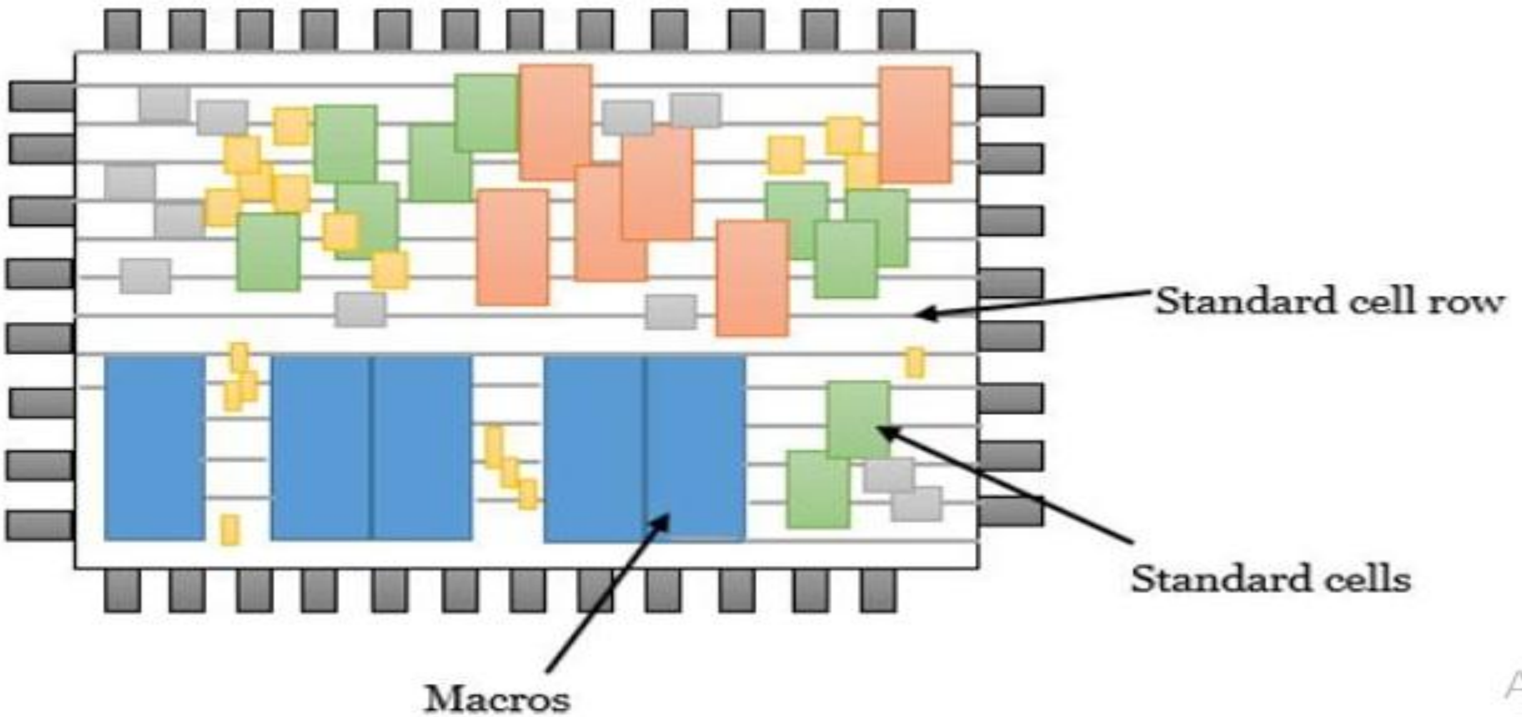


Fig : Coarse placement of standard cells

Detailed or Legal Placement

Legalization:

- During legalization, the tool moves the cells to legal locations on the placement grid and eliminate any overlap between cells.
- These small changes to cell location cause the lengths of the wire connections to change, possibly causing new timing violations.
- Such violations can often be fixed by incremental optimization, for example: by resizing the driving cells.

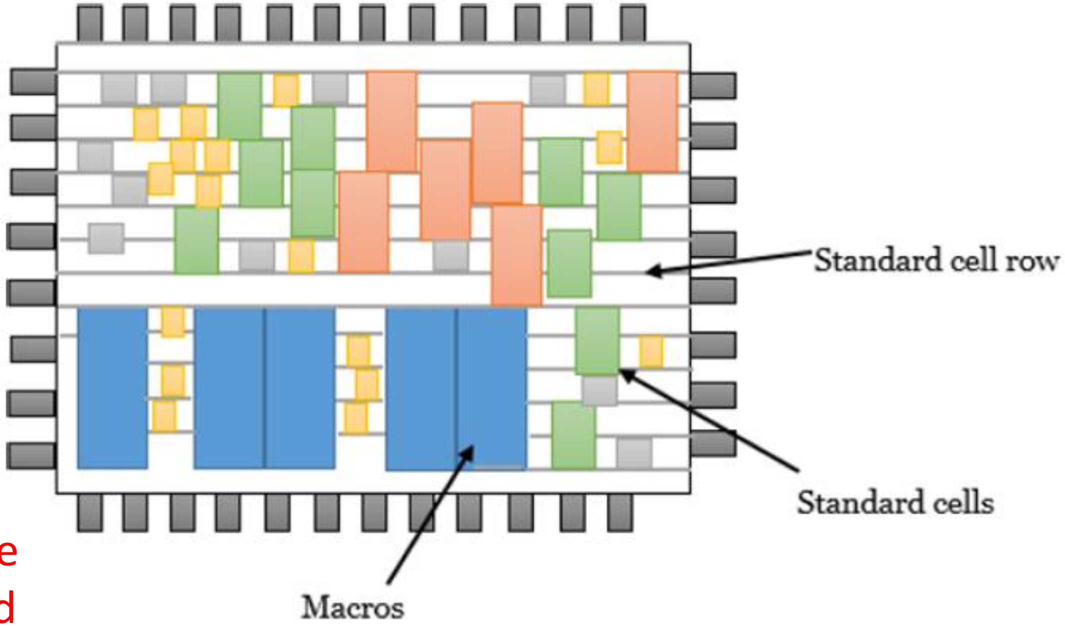
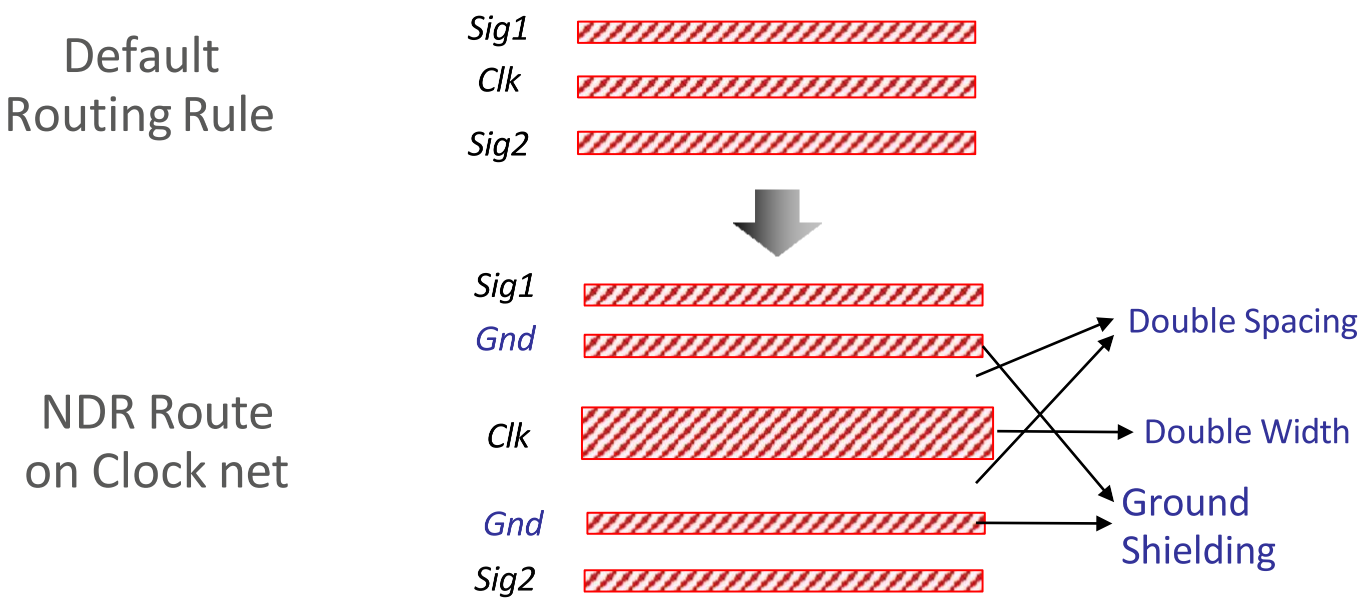


Fig : Legalization of standard cells

After Placement, generate Congestion, Utilization and Timing Report

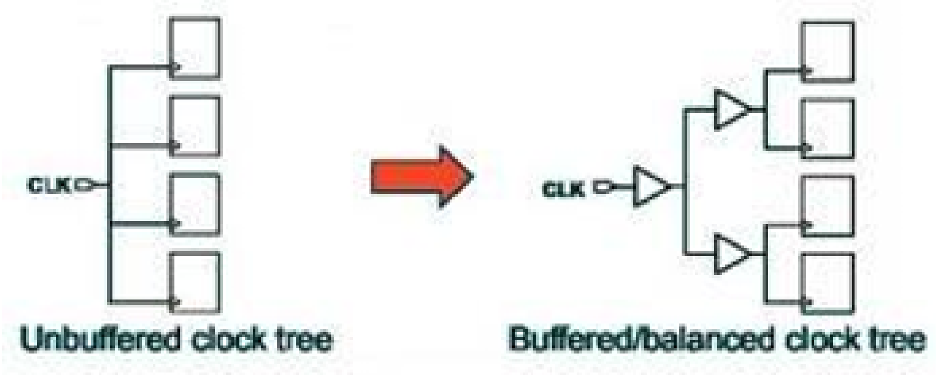
Pre CTS Optimization

- Non-Default Rule (NDR)
  - The user-defined Routing rules apart from the default Routing Rule
  - Often used to “harden” the sensitive nets like Clock Nets
  - NDRs make the Clock Routes less sensitive to CrossTalk or EM effects
  - Double/ Triple Width for avoiding Electromigration
  - Double/ Triple Spacing for avoiding Crosstalk
  - NDRs will improve Insertion Delay



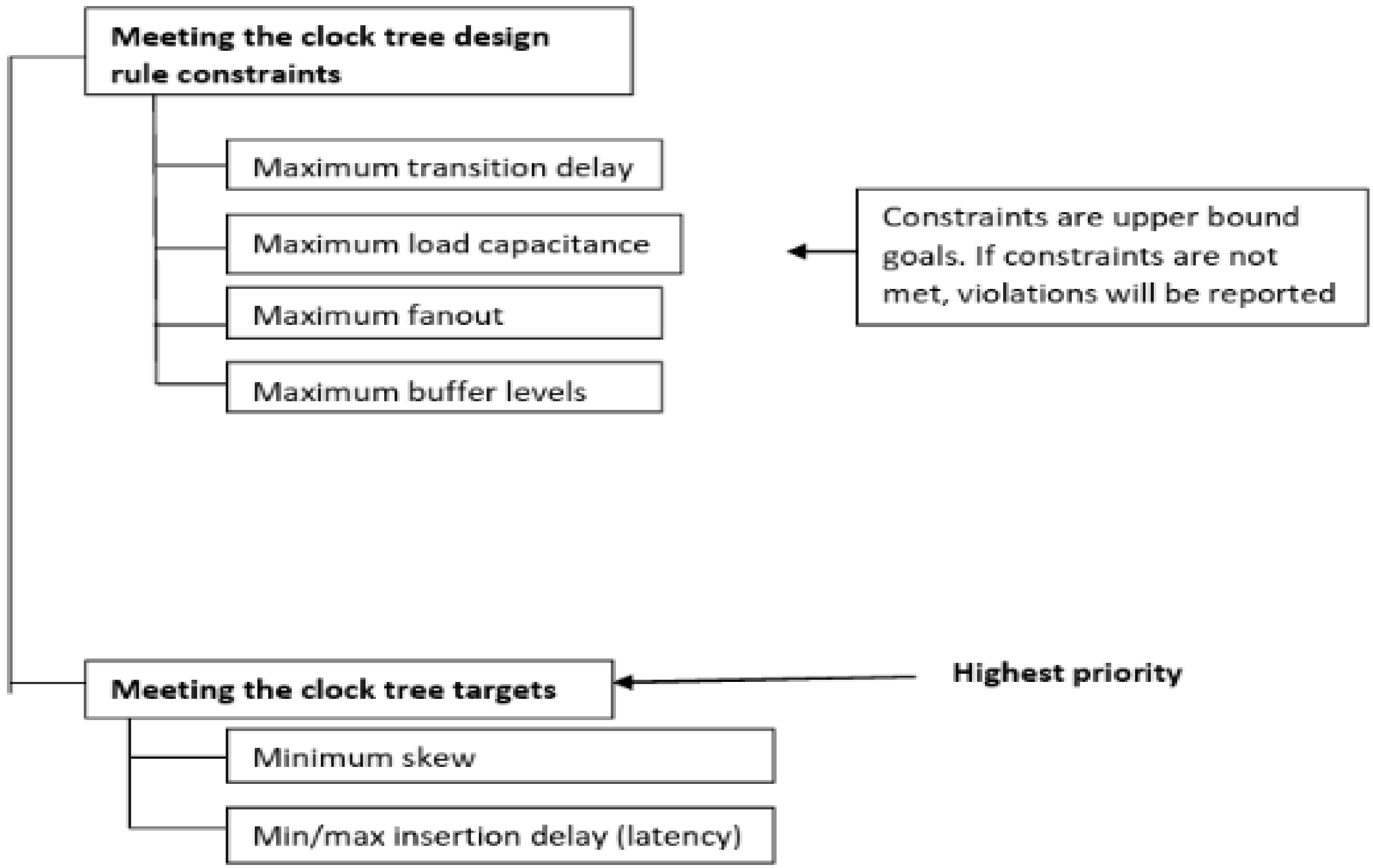
CLOCK-TREE SYNTHESIS

- CTS is one of the most important stages in PnR. CTS QOR decides timing convergence and power. In most of the ICs clock consumes 30 to 40% of total power. So efficient clock architecture, clock gating and clock tree implementation helps to reduce power
- The process of distributing the clock and balancing the load is called CTS. Basically, delivering the clock to all sequential elements. CTS is the process of insertion of buffers and inverters along the clock paths of ASIC design in order to achieve zero or minimum skew or balanced skew. Before CTS, all clock pins are driven by a single clock source.CTS starting point is clock source and CTS ending point is clock pin of sequential cells.



CLOCK-TREE SYNTHESIS

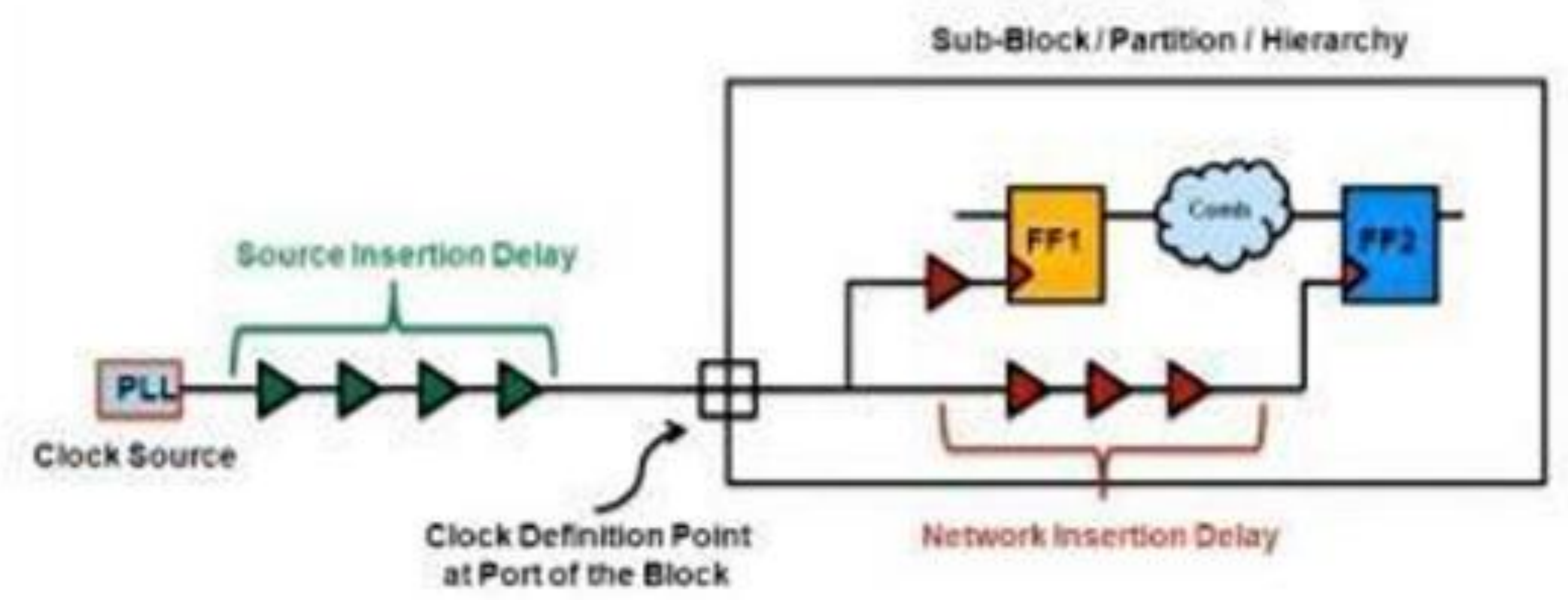
Goals of CTS



CLOCK-TREE SYNTHESIS

Different Clock Terms:

a) Clock latency/insertion delay

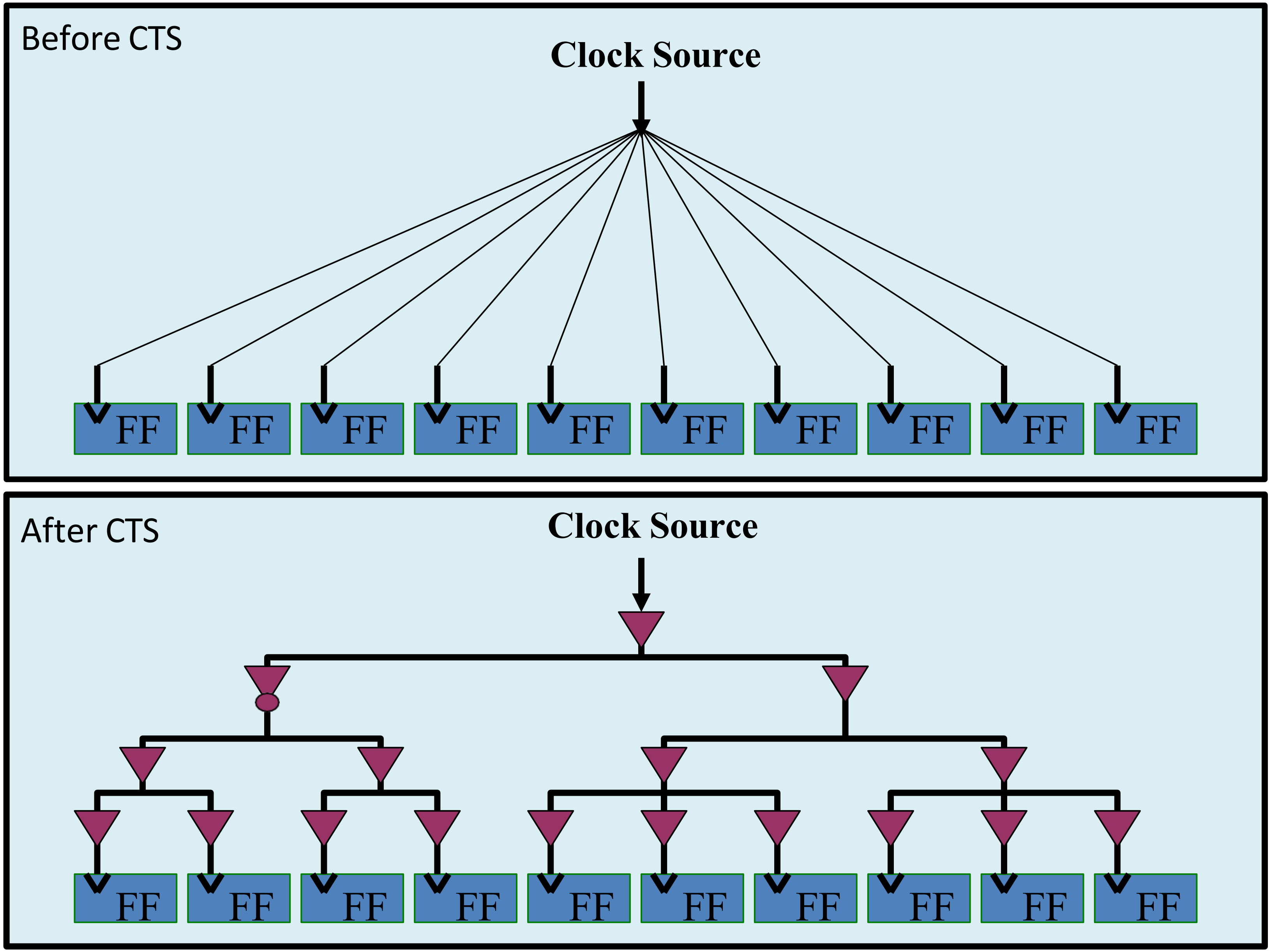


- Total time taken by the clock signal to reach the inputs of the register
- Source latency is the time between clock sources to clock definition ports
- Network latency is the time between clock definition ports to clock leaf cells in the design

Clock Tree Synthesis (CTS)

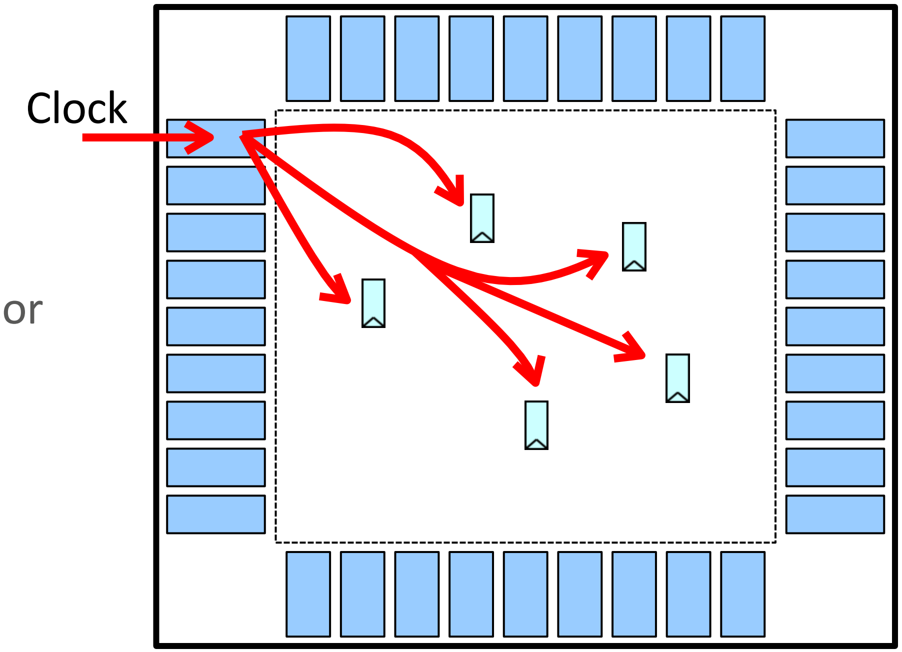
- The Clock Problem
  - Clock skew
  - Long clock insertion delay
  - Skew across clocks
  - Heavy clock net loading
  - Clock is power hungry
  - Clock to signal coupling effect (CrossTalk)
  - Electromigration on clock net
- Clock Tree is a path from the Clock Source (Root) to Clock Sinks (Leaf)
- Clock Tree Synthesis is the process of creating this Clock Path from Clock Source to Clock Sinks
- All Clock pins of flip Flop are considered as Clock Sinks (Leaf); where the Clock Tree Synthesis ends

Clock Tree Synthesis (CTS)



Clock Tree Synthesis (CTS)

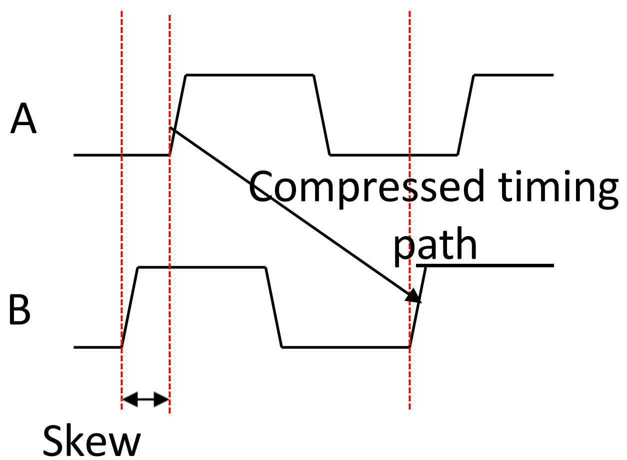
- Main concerns for Clock Design
  - Skew
    - Most important concern for clock networks
    - For increased clock frequency, skew may contribute over 10% of the system cycle time
    - Due to variations in trace length, metal width and height, coupling caps
    - It can also be due to variations in local clock load, local power supply, local gate length and threshold, local temperature
  - Power
    - Very important, as clock is a major power consumer
    - It switches at every clock cycle
  - Noise
    - Clock is often a very strong aggressor
    - May need shielding
  - Delay
    - Not really important
    - But Slew Rate is important (sharp transition)

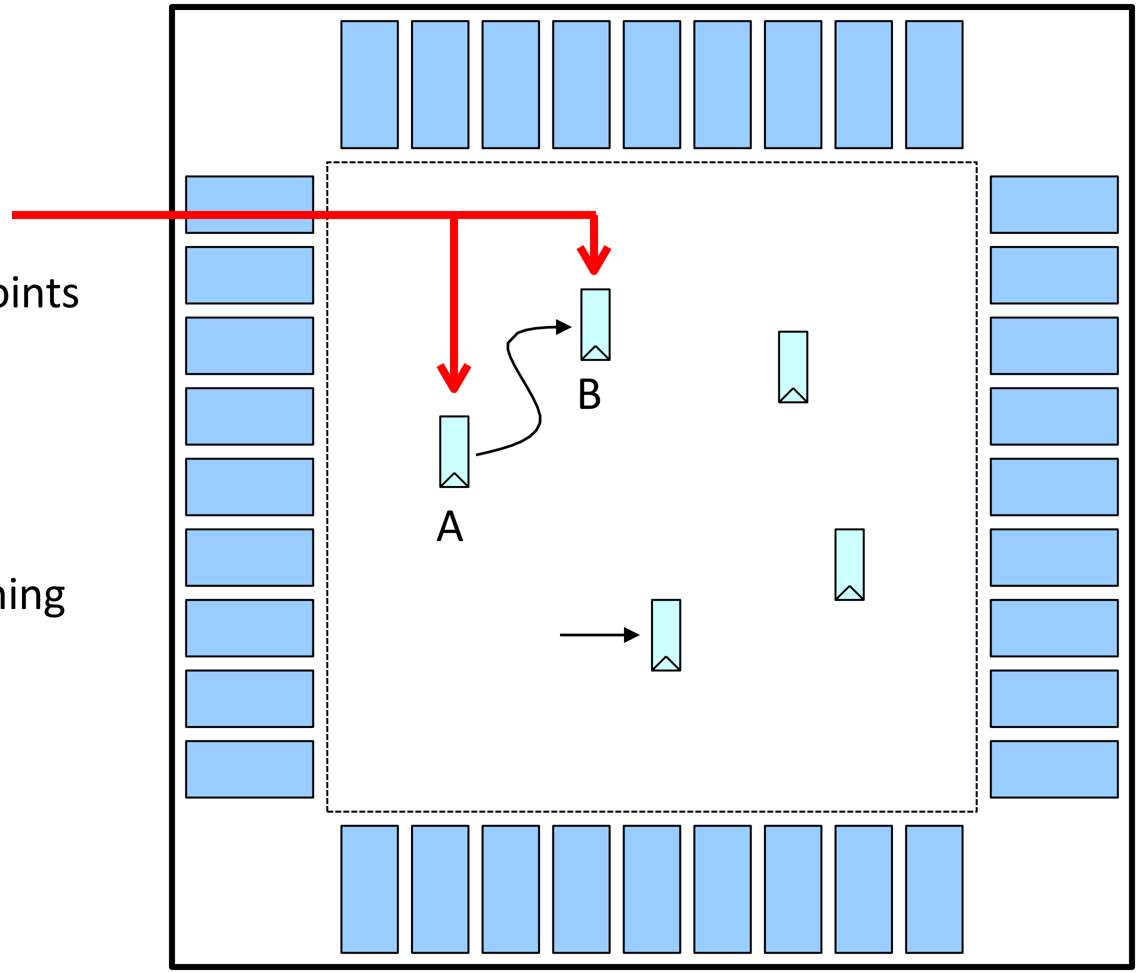


Clock Tree Synthesis (CTS)

- Clock Skew: Spatial Clock Variation

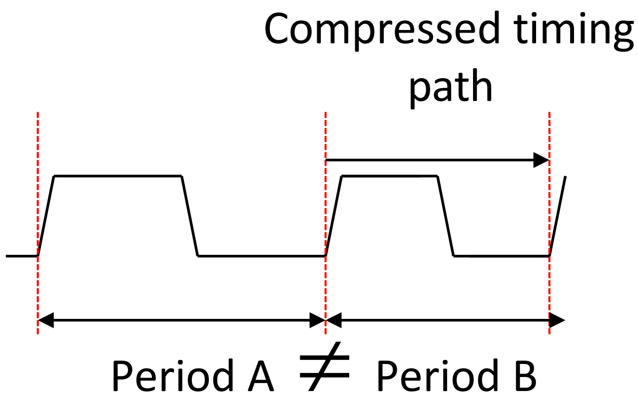
Clock Skew
Difference in clock arrival time at two spatially distinct points



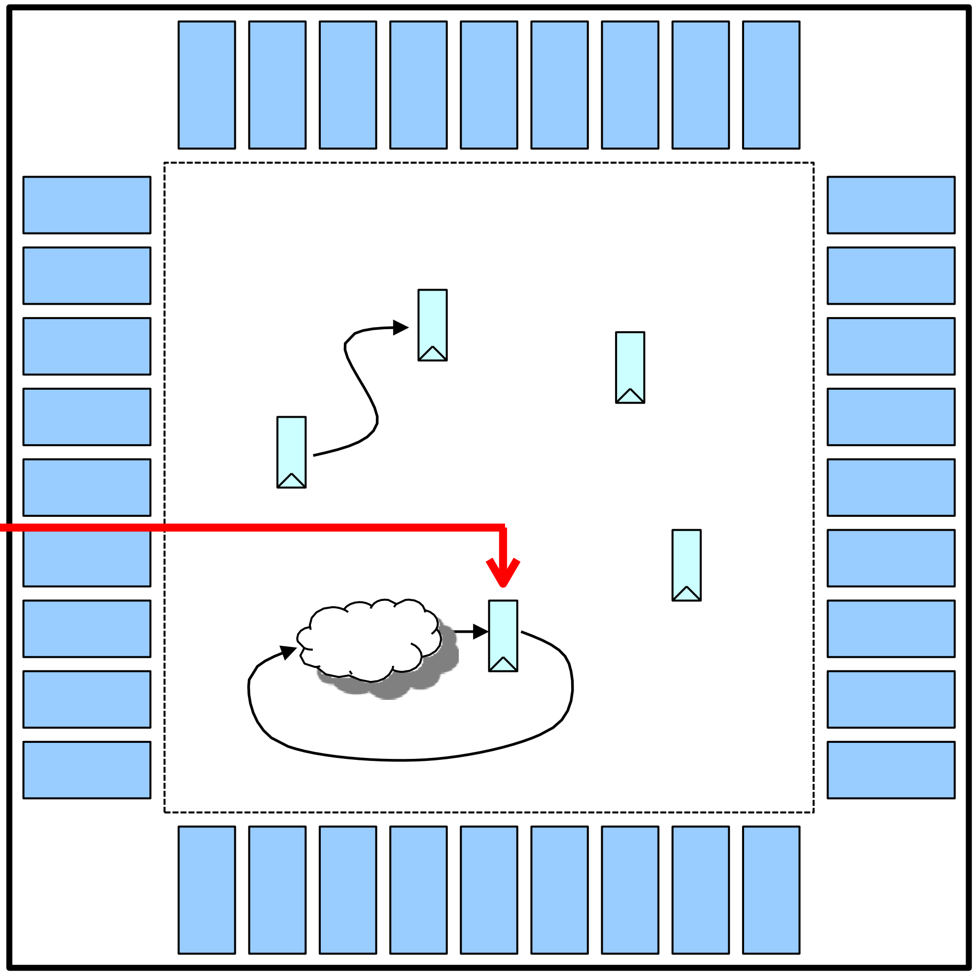


Clock Tree Synthesis (CTS)

- Clock Jitter: Temporal Clock Variation



Clock Jitter
Difference in clock period over time



Clock Tree Synthesis (CTS)

- CTS Pre-requisites
  - Legally Placed and Optimized with acceptable Congestion
  - Timing should be good
  - No Design Rule Violations
  - Power/Ground nets are pre-routed
  - HFNS done
  - Logical/Physical Library should have special Clock Cells
- CTS Objects
  - The timer starts from every Clock Source and traces forward over Combinational Arcs until it reaches the Clock Pin of a flop or another Clock Source
  - All Pins/ Timing Arcs in the forward trace before a valid Leaf are considered to be in the clock network
  - Pin or Combinational Timing Arcs that trace to a non-clock pin are not part of Clock Tree network (e.g. D pin of FF)
  - Sequential elements are traced through if it is a source of the Generated Clock
  - Clock tracing after the propagation of Case Analysis
  - Clock tracing should be Mode aware
  - Inverters are added in Clock Tree for better Duty Cycle

Clock Tree Synthesis (CTS)

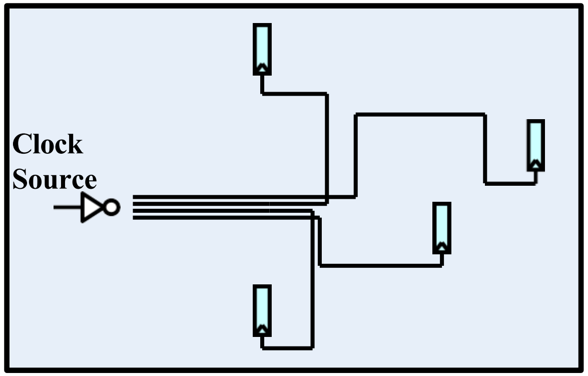
- CTS Flow
  - Check and fix Macro locations
  - Read CTS SDC: Clock Tree begins at SDC defined clock pin and ends at stop pin of the flop
  - Generate CTS Specification file
    - Max. Skew
    - Max. and Min. Insertion Delay
    - Max. Transition, Capacitance, Fanout
    - No. Buffer levels (Tree depth)
    - Buffer/ Inverter list
    - Clock Tree Routing Metal Layers
    - Clock Tree Leaf Pin, Root Pin, Preserve Pin, Through Pin and Exclude Pin
  - Compile CTS using CTS Spec. file
  - Place Clock Tree Cells
  - Route Clock Tree (Optional and can be done during Signal net

Example of CTS spec file

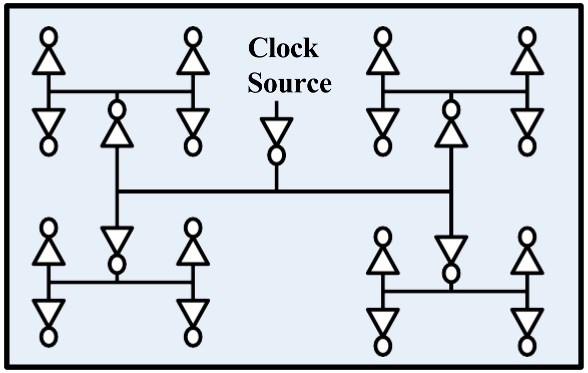
```
AutoCTSRootPin  SH1/I23/Z
ExcludePin + XPU/CAM/C
MaxDelay  5ns
MinDelay  0ns
Buffer buf1 buf2 inv1 inv2 del1
MaxSkew  500ps
MaxDepth 20
LeafPin + FPU/CORE/A rising
END
```

Clock Tree Synthesis (CTS)

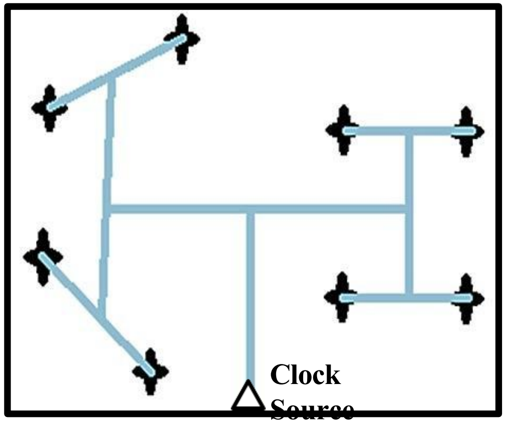
- CTS Algorithms
  - RC Tree Based CTS
  - H Tree based Algorithm
  - X Tree based Algorithm
  - Method of Mean and Median (MMM)
  - Geometric Matching Algorithm (GMA)
  - Pi Configuration



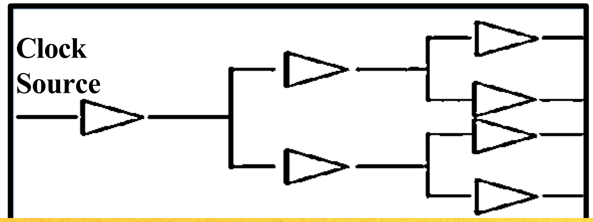
RC-Tree



H-Tree

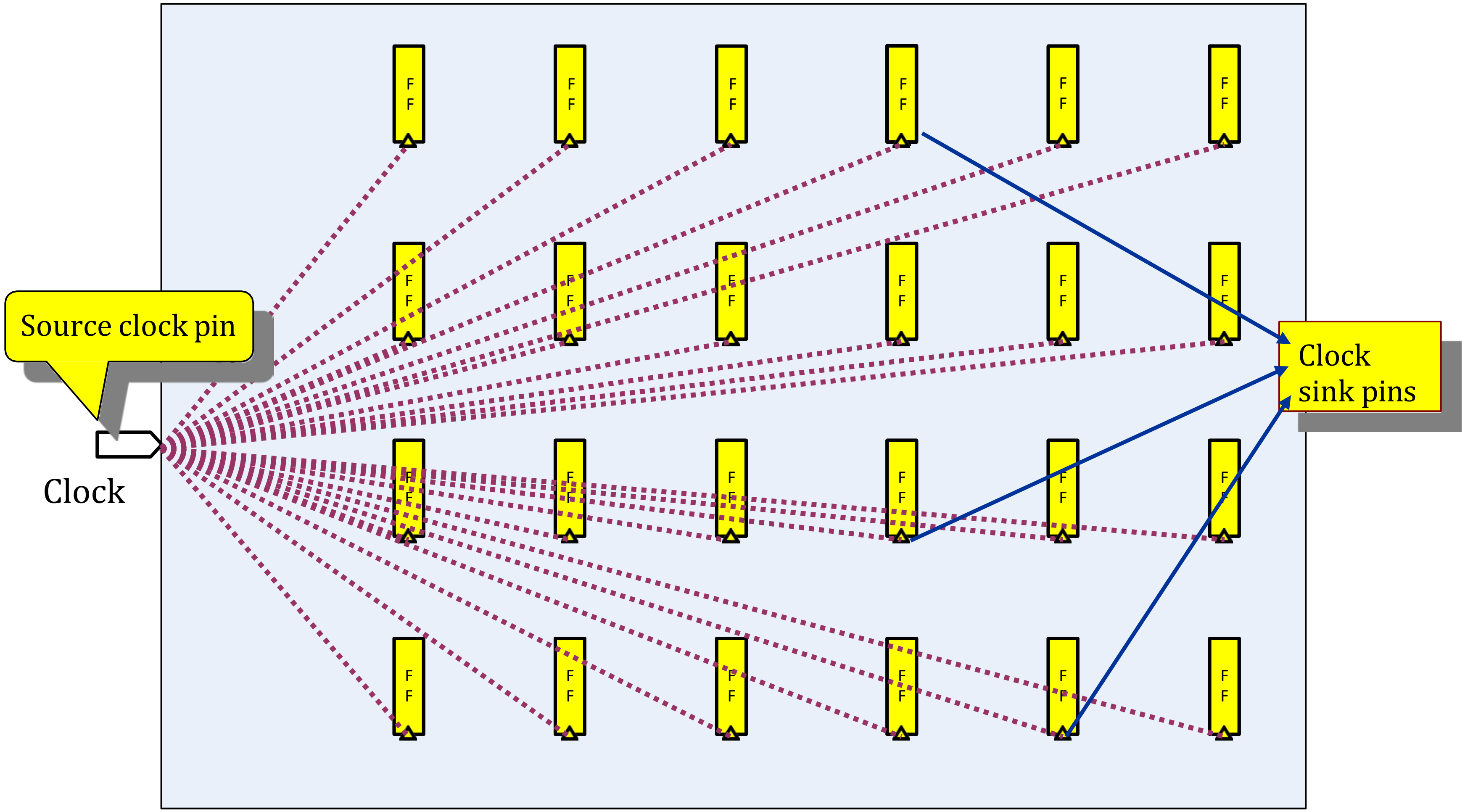


GMA



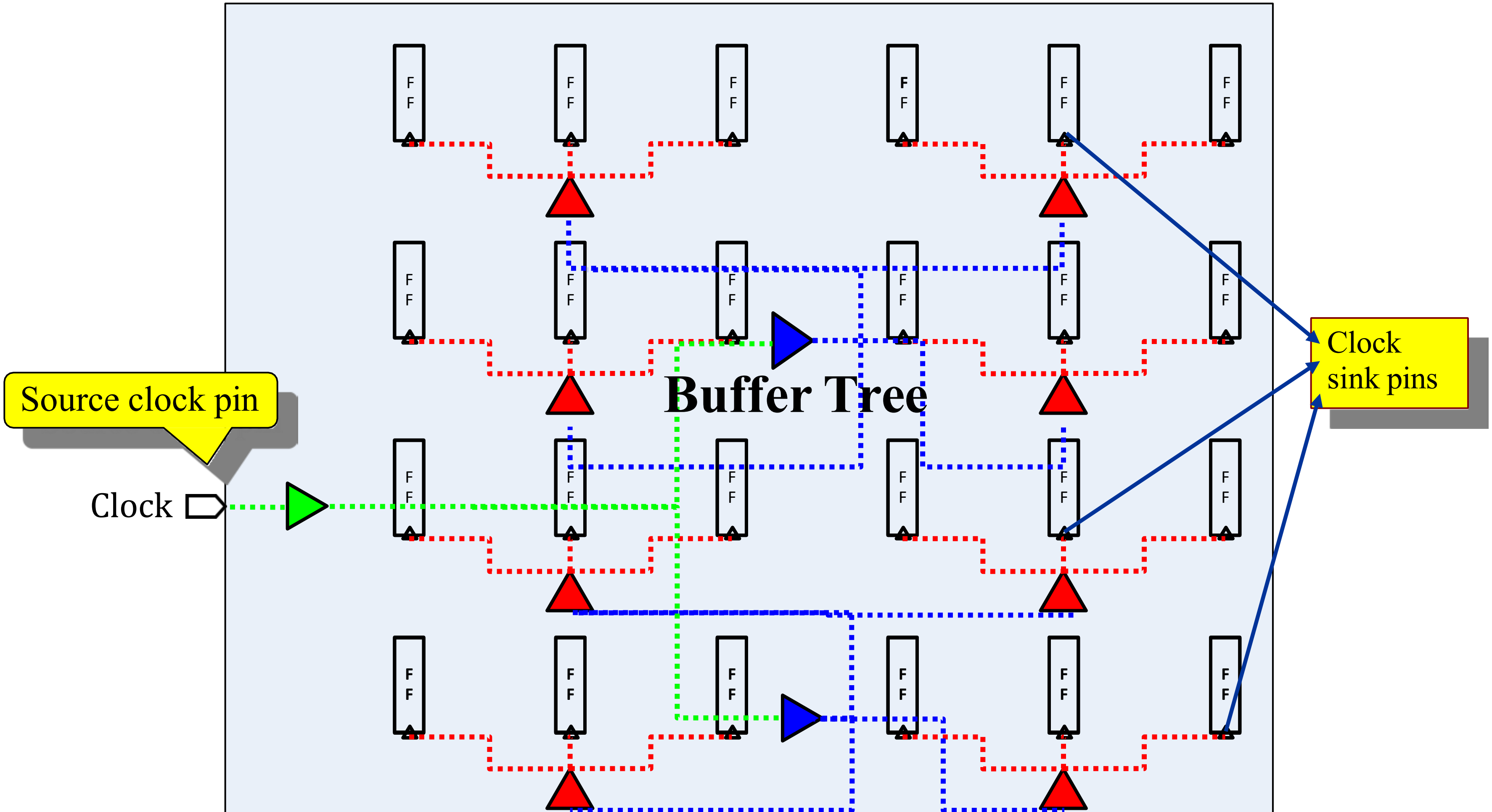
Clock Tree Synthesis (CTS)

- Before CTS all Clock Pins are driven by a single Clock Source



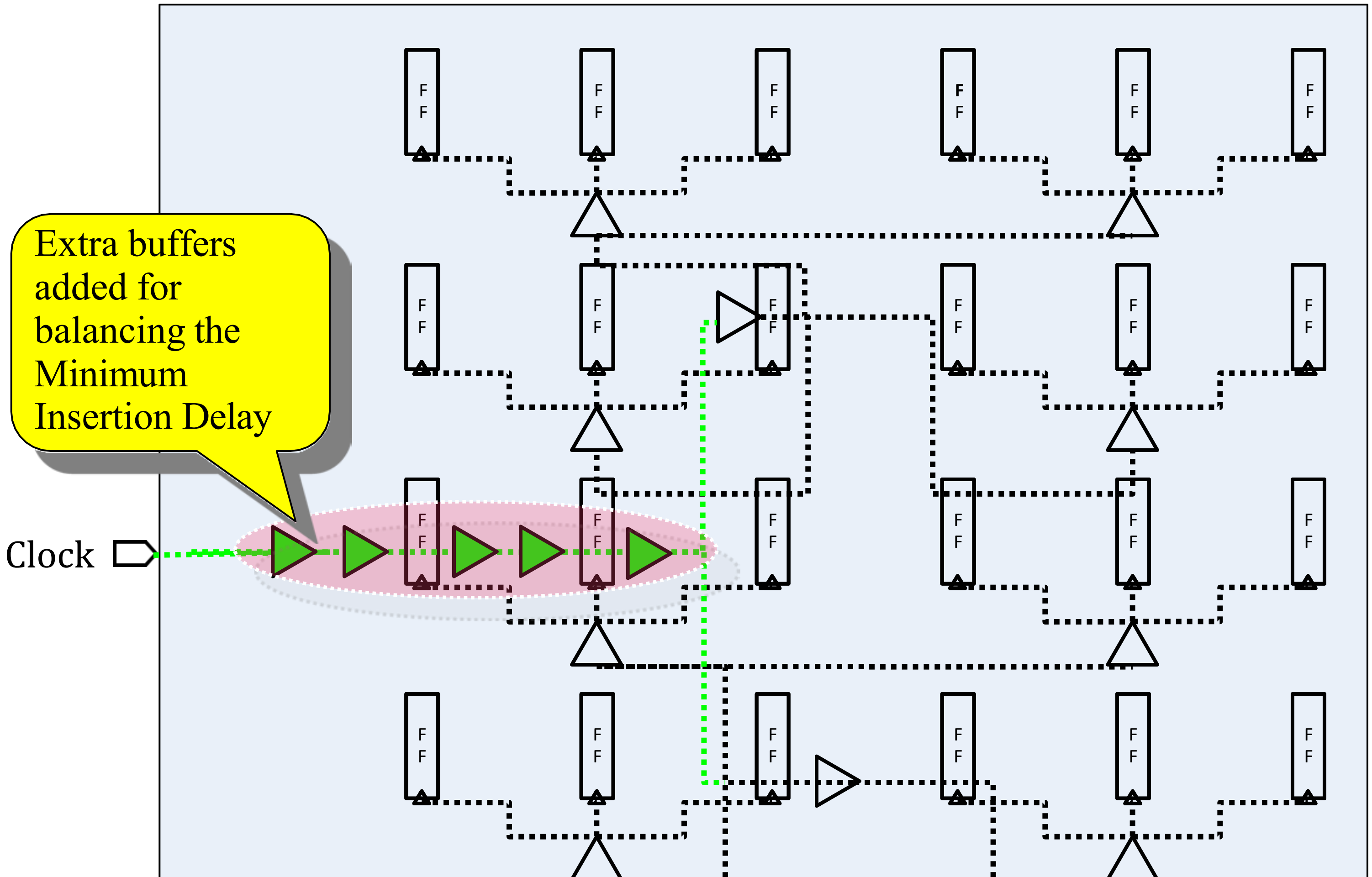
Clock Tree Synthesis (CTS)

- After CTS the buffer tree is built to balance the loads and minimize the skew



Clock Tree Synthesis (CTS)

- After CTS a "delay line" is added to meet the minimum Insertion Delay (ID)



Clock Tree Synthesis (CTS)

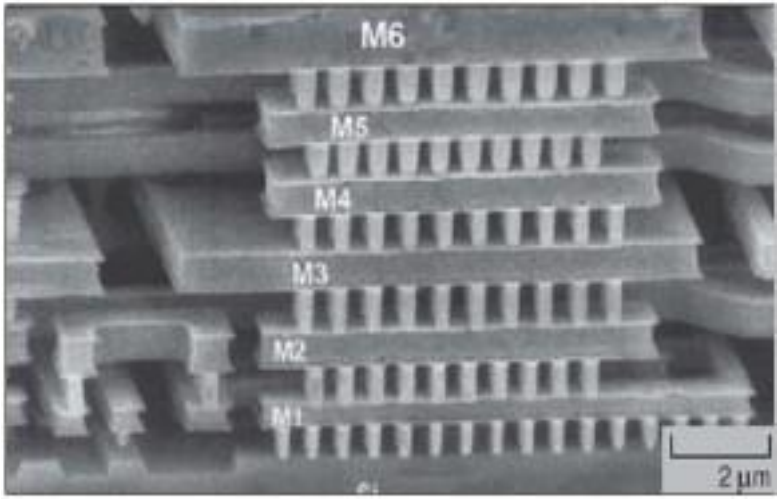
- Analyze the Clock Tree
  - Report Timing (both Setup and Hold)
  - If timing not met then check clocks be grouped (balanced together)
  - Report Insertion Delay & Skew and verify that the targets are achieved
  - Report DRV targets (Fanout, Capacitance and Transition)
  - Check the intended Leaf Cell (Clock Sinks) is reached
  - Check the Clock Tree Exceptions are not in the Clock Tree
  - Report the pre-existing cells, such as Clock Gating Cells
  - Do Quality-of-Report (QoR)
  - Check Clock Tree converges either with itself or with another Clock Tree
  - Clock Tree has timing relationship with other Clock Trees for inter Clock Skew balancing
  - Check Design Rule Constraints
  - Check Routing Constraints
  - Report Power and Area

Post CTS Optimization

- Post CTS Optimization
  - Optimization with Useful Skew
  - Optimization with Total Negative Slack (TNS)
  - Fine Grid Spacing
  - Post CTS Optimization Techniques
    - Shielding
    - Sizing
    - Buffer re-location
    - Level adjustment
  - Optimize the design for Hold Time
    - Hold Violations should be fixed first in Best Corner and then in Worst Corner
  - Area Optimizations

Routing

- Importance of Routing as Technology shrinks
  - Device (Gate) delay decreases
  - Interconnect resistance increases
  - Vertical heights of interconnect layers increase, in an attempt to offset increasing interconnect resistance
  - Area component of interconnect capacitance no longer dominates
  - Lateral (sidewall) and fringing components of capacitance start to dominate the total capacitance of the interconnect
  - Interconnect capacitance dominates total Gate loading

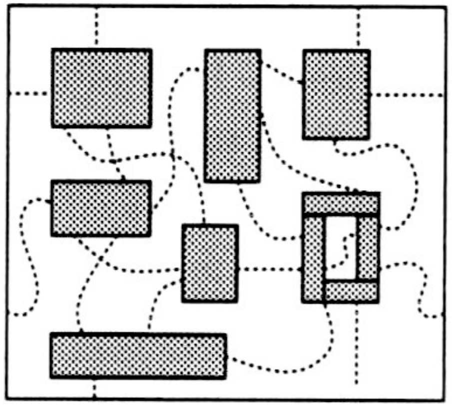


Multi-level Interconnection (MLI) Technology Layer stacks

- Routing Objectives
  - Skew requirements
  - Open/Short circuit clean
  - Routed paths must meet setup and hold timing margin
  - DRVs max. Capacitance/ Transition must be under the limit
  - Metal traces must meet foundry physical DRC requirements

Routing

- Routing Stages
  - Trial/Global Routing
    - Identifying routable path for the nets driving/ driven pins in a shortest distance
    - Does not consider DRC rules, which gives an overall view of routing and congested nets
    - Assign layers to the nets
    - Identify and assign net segments over the specific routable window called Global Route Cell (GRC)
    - Avoid congested areas and also long detours
    - Avoid routing over blockages
    - Avoid routing for pre-route nets such as Rings/Stripes/Rails
    - Uses Steiner Tree and Maze algorithm

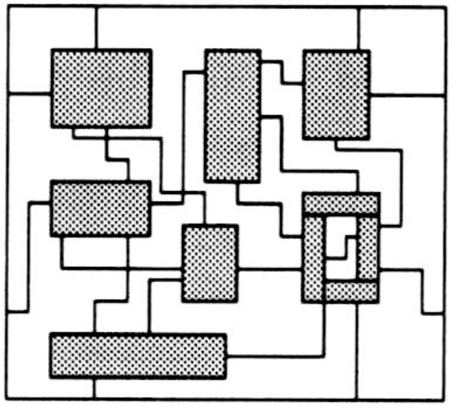


Global Routing

  - Track Assignment
    - Takes the Global Routed Layout and assigns each nets to the specific Tracks and layer geometry
    - It does not follow the physical DRC rules
    - It will do the timing aware Track Assignment
    - It helps in Via Minimization

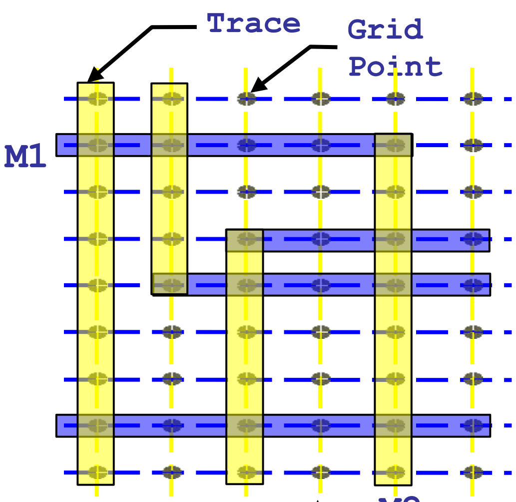
Routing

- Routing Stages
  - Detail/Nano Routing
    - Detailed routing follows up with the track routed net segments and performs the complete DRC aware and timing driven routing
    - It is the final routing for the design built after the CTS and the timing is freeze
    - Filler Cells are adding before Detailed Routing
    - Detail Routing is done after analyze the cause for congestion in the design, add density screen or change flooplan etc.



Detailed Routing

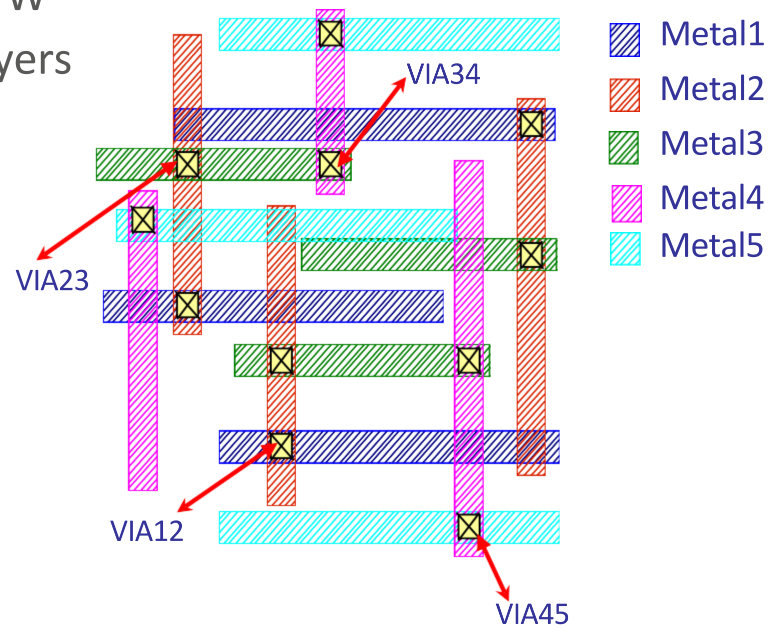
- Grid Based Routing
  - Metal traces (routes) are built along and centered upon routing tracks on the grid points
  - Various types of grids are Manufacturing Grid, Routing Grid (Pitch) and Placement Grid
  - Grid dimension should be multiple of Manufacturing Grid



Routing

- Routing Preferences
  - Typically Routing only in "Manhattan" N/S E/W directions

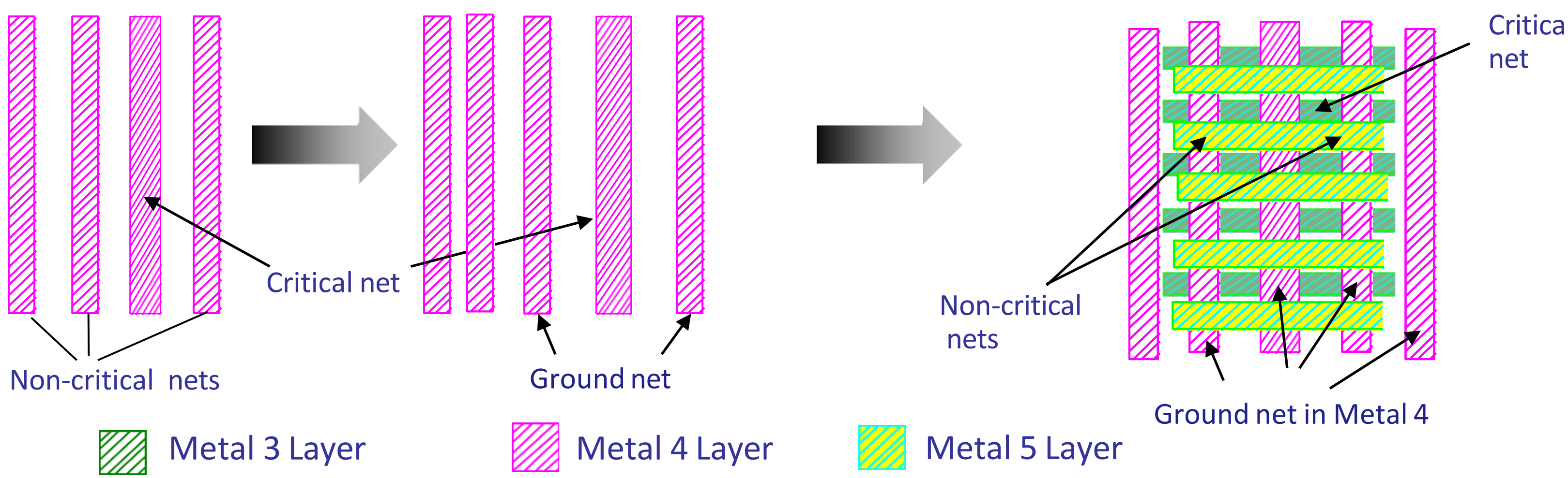
    E.g. layer 1 – N/S Layer 2 – E/W
  - Spacing checks with the adjacent layers
  - Width checks for all layers
  - Via dimension rules
  - Slotting rules
  - A segment cannot cross another segment on the same wiring layer
  - Wire segments can cross wires on other layers
  - Power and Ground have their own layers, mostly the top layers



- Layer Routing directions: Each metal layer has its own preferred routing direction and are defined in a technology rule file
  - M1: Horizontal, M2: Vertical , M3: Horizontal, M4: Vertical and so on
- In some cases, we can avoid following preferred routing direction for smart routing (Non-preferred direction)

Post Routing Optimization

- Signal Integrity (SI) Optimization by NDRs and Shielding for the sensitive nets
- Types of Shielding for sensitive nets
  - Same layer shielding
  - Adjacent layer/ Coaxial shielding



Same Layer shielding

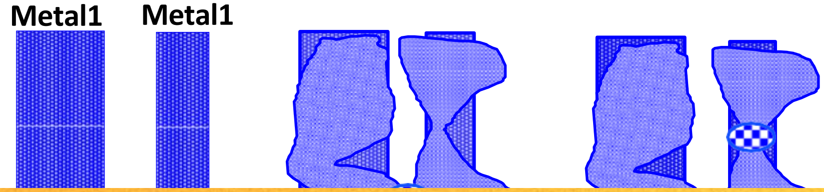
Adjacent Layer/ Coaxial shielding

Physical Verification (DRC)

- Design Rule Check (DRC) is the process of checking physical layout data against fabrication-specific rules specified by the foundry to ensure successful fabrication
- Process specific design rules must be followed when drawing layouts to avoid any manufacturing defects during the fabrication of an IC
- Process design rules are the minimum allowable drawing dimensions which affects the X and Y dimensions of layout and not the depth/vertical dimensions
- As Technology Shrinks
  - Number of Design Rules are increasing
  - Complexity of Routing Rules is increasing
  - Increasing the number of objects involved
  - More Design Rules depending on Width, Halo, Parallel Length

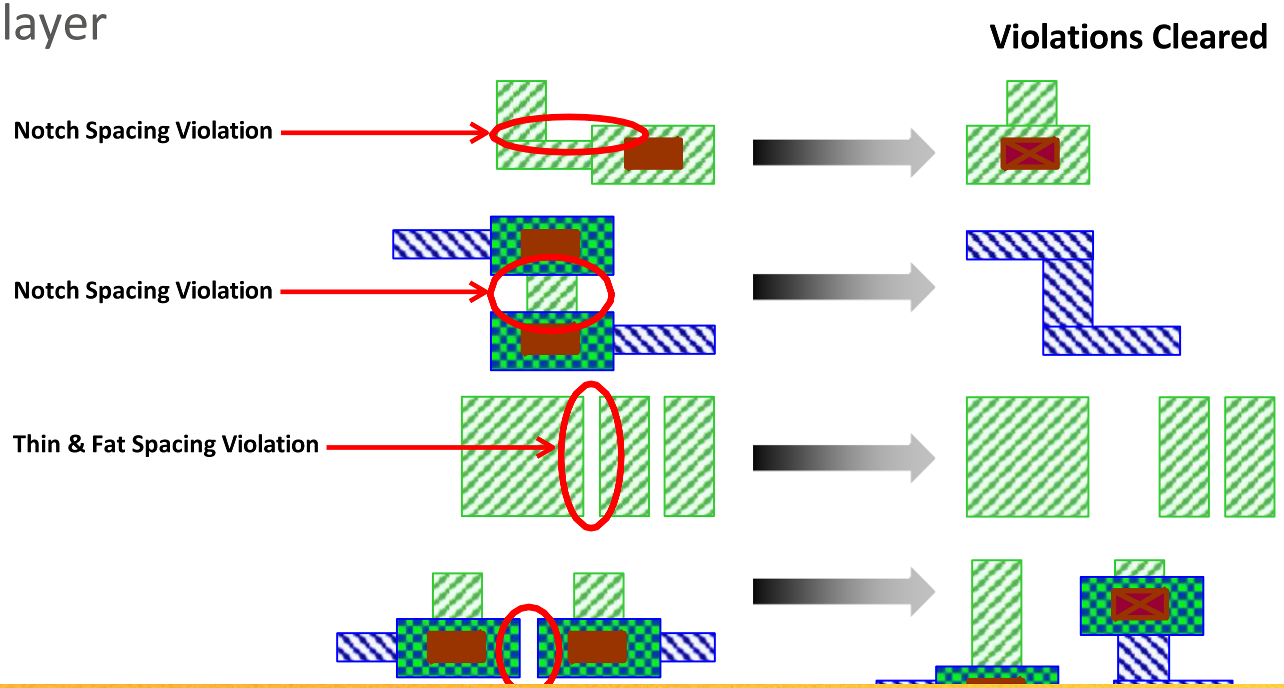
DRC Rule	130nm	90nm	65nm	45nm
Width-based Spacing	1-2	2-3	3-5	7
Min-Area Rule	1 pitch	2 pitch	3 pitch	5 pitch
Cut Number (Via)	N/A	1-2	4-5	5-6
Dense EoL (OPC)	N/A	N/A	M1/M2	All Layers
Min-step (OPC)	N/A	1	5	5

- Violating a design rule might result in a non-functional circuit or low Yield



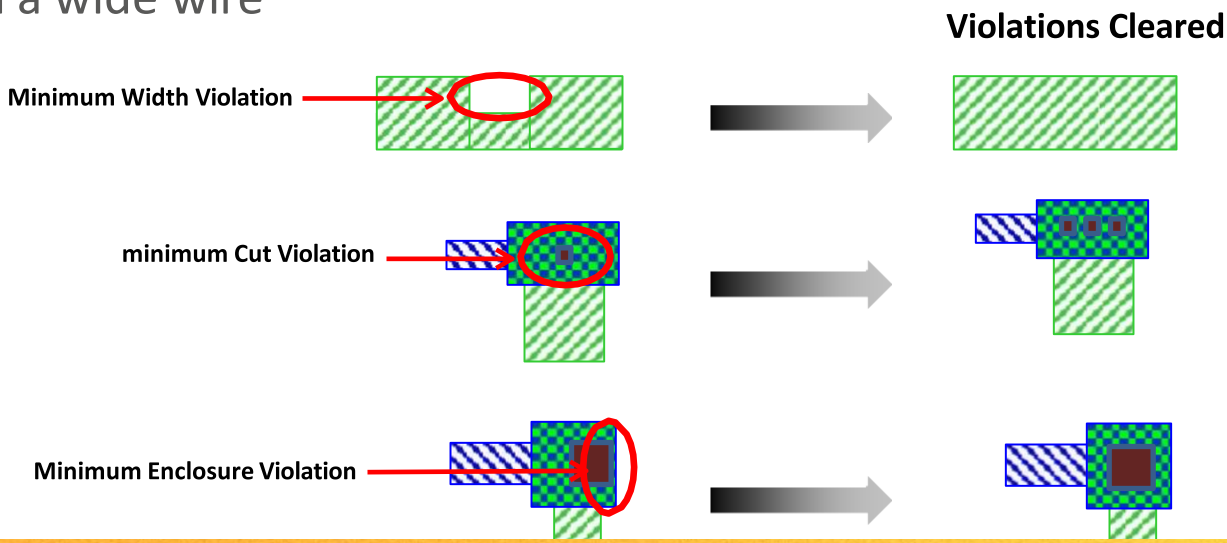
Physical Verification (DRC)

- Design Rule examples
  - Maximum Rules: Manufacturing of large continuous regions can lead to stress cracks. So ‘wide metal’ must be ‘slotted’ (holes)
  - Angles: Usually only multiples of 45 degree are allowed
  - Grid: All corner points must lie on a minimal grid, otherwise an “off grid error” is produced
  - Minimum Spacing: The minimum spacing between objects on a single layer



Physical Verification (DRC)

- Design Rule examples
  - Minimum Width: The min width rule specifies the minimum width of individual shapes on a single layer
  - Minimum Enclosure/ Overlap: Implies that the second layer is fully enclosed by the first one
  - Notch: The rule specifies the minimum spacing rule for objects on the same net, including defining the minimum notch on a single-layer, merged object
  - Minimum Cut: the minimum number of cuts a via must have when it is on a wide wire

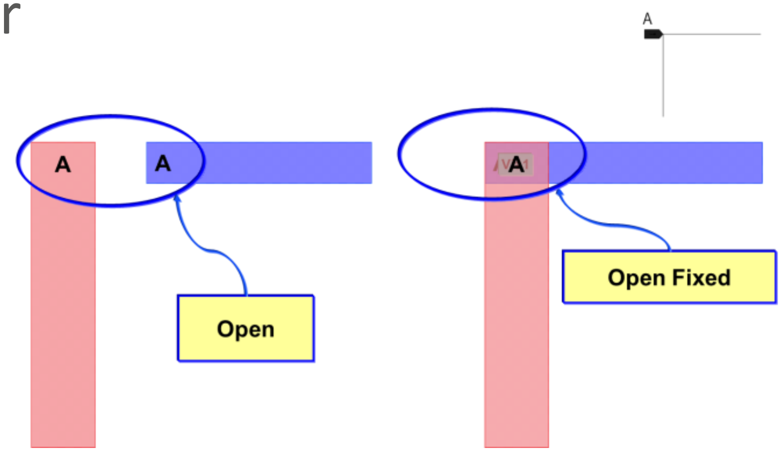


Physical Verification (LVS)

- Layout Versus Schematic (LVS) verifies the connectivity of a Verilog Netlist and Layout Netlist (Extracted Netlist from GDS)
- Tool extracts circuit devices and interconnects from the layout and saved as Layout Netlist (SPICE format)
- As LVS performs comparison between 2 Netlist, it does not compare the functionalities of both the Netlist
- Input Requirements
  - LVS Rule deck
  - Verilog Netlist
  - Physical layout database (GDS)
  - Spice Netlist (Extracted by the tool from GDS)
- LVS checks examples
  - Short Net Error, Open Net Error, Extract errors, Compare errors

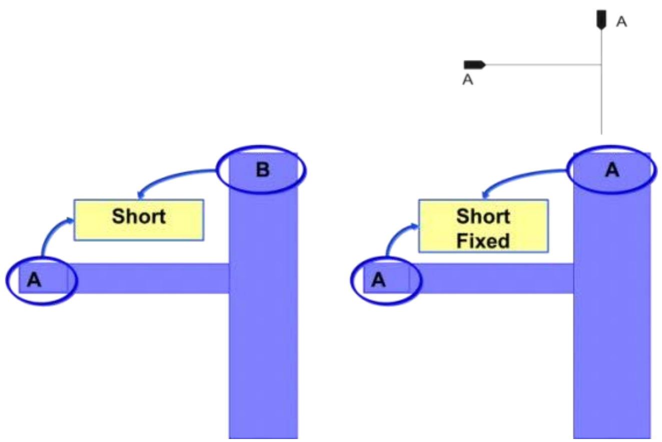
Physical Verification (LVS)

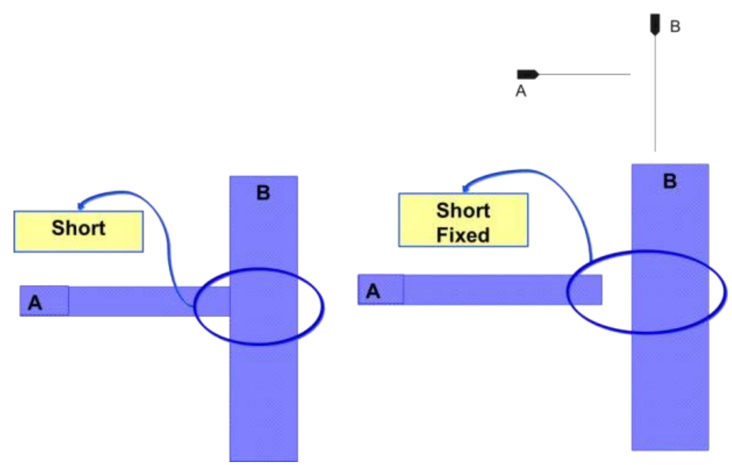
- Open Net Error



Same net is routed in two different metal layers but not connected

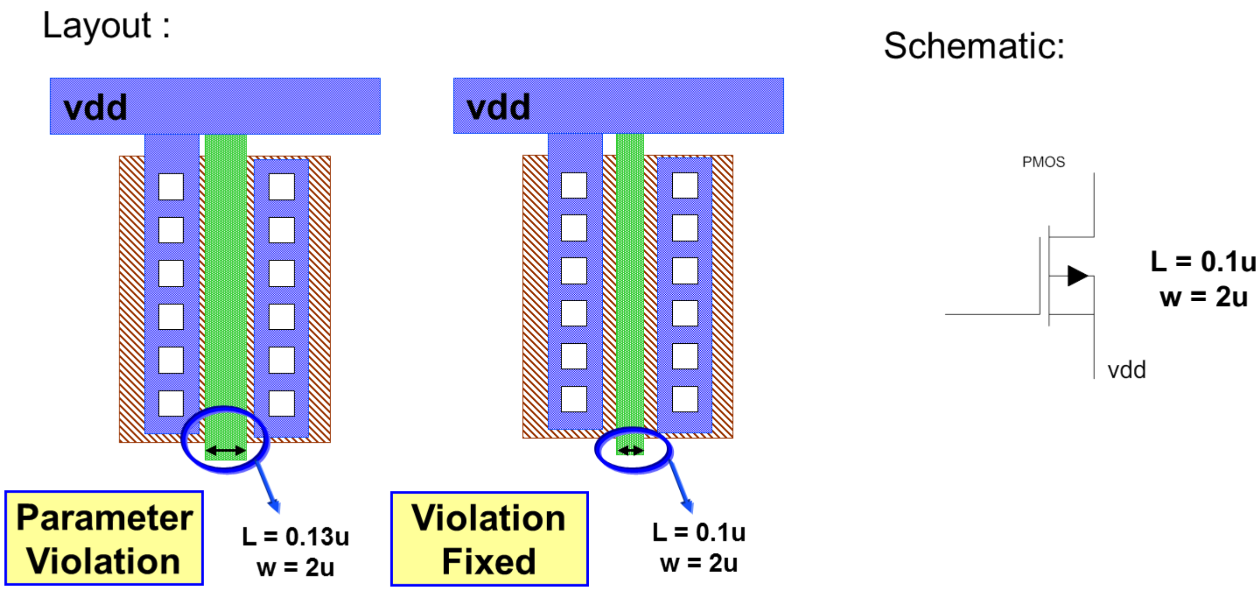
- Short Net Error





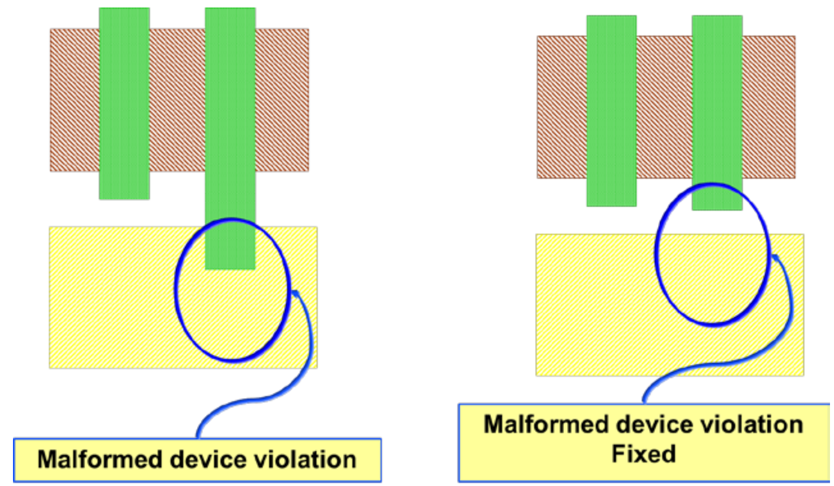
Physical Verification (LVS)

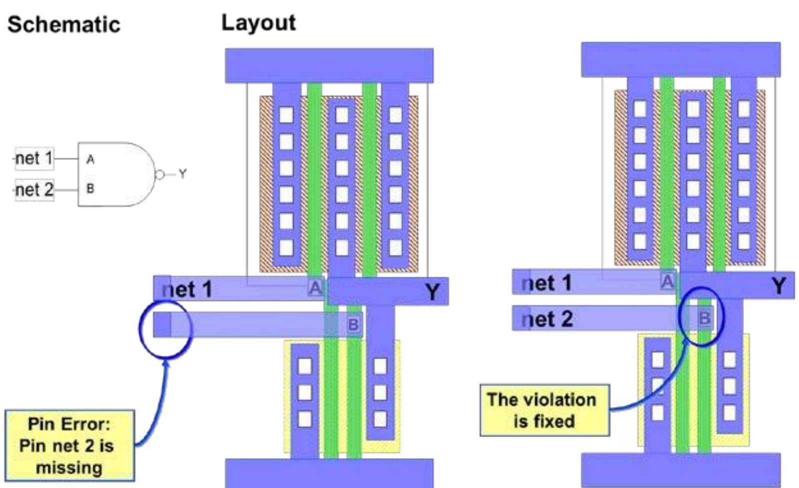
- Extract Errors
  - Parameter Mismatch
  - Device parameters on schematic and layout are compared
  - Example: Let us consider a transistor here, LVS checks are necessary parameters like width, length, multiplication factor etc.

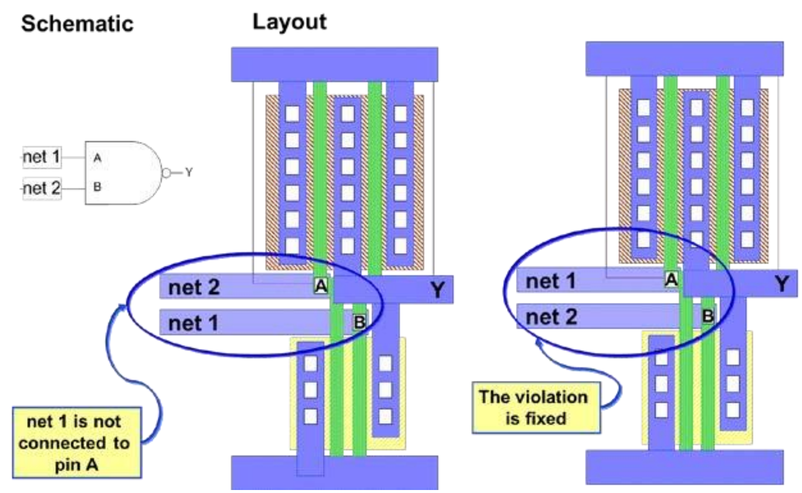


Physical Verification (LVS)

- Compare Errors
  - Malformed Devices
  - Pin Errors
  - Device Mismatch
  - Net Mismatch

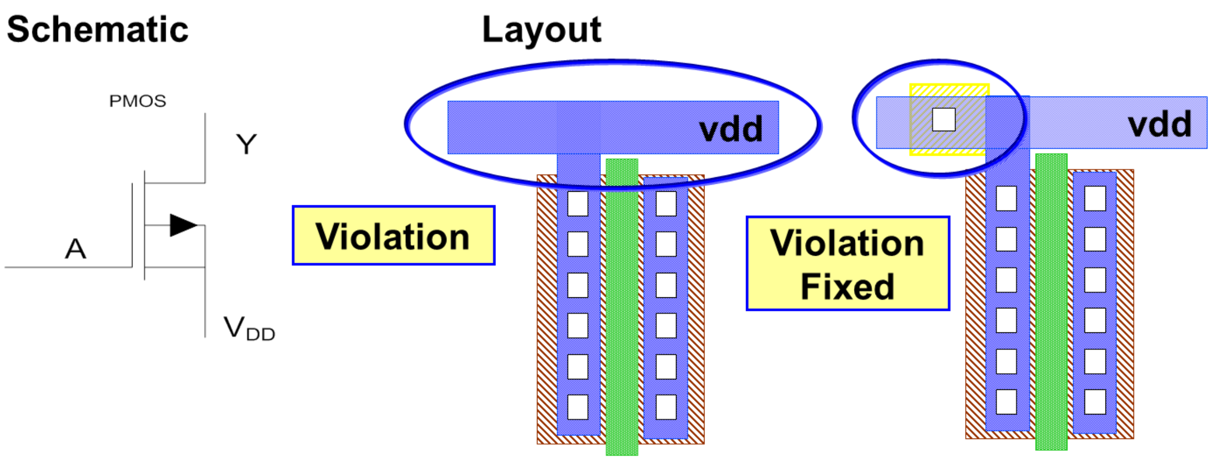






Physical Verification (ERC)

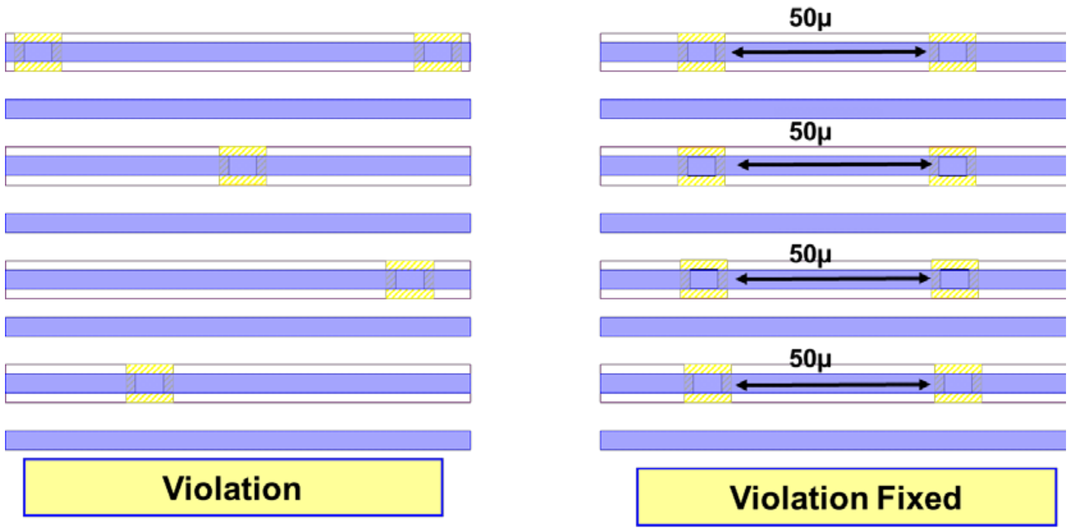
- Electrical Rule Check (ERC) is used to analyze or confirm the electrical connectivity of an IC design
- ERC checks are run to identify the following errors in layout
  - To locate devices connected directly between Power and Ground
  - To locate floating Devices, Substrates and Wells
  - To locate devices which are shorted
  - To locate devices with missing connections
- Well Tap connection error: The Well Taps should bias the Wells as specified in the schematics



Physical Verification (ERC)

- Well Tap Density Error: If there is no enough Taps for a given area then this error is flagged
- Taps need to be placed regularly which biases the Well to prevent Latch-up

  e.g., In typical 90nm process the Well Tap Density Rule require Well-taps to be placed every 50 microns
- Tools: Mentor Graphics Calibre, Synopsys Hercules, Cadence Assura, Magma Quartz



DFM Checks

- Antenna Check (Gate-Oxide Integrity check)
  - Maximum net length restriction connected to Gate terminal
- Redundant Contacts/ Via
  - Multiple Via improves both Yield and Timing by resistance paralleling
- Metal Filling
  - Narrow Metal Layer separated from other Metal Layers may get high density of etchant than closely spaced wires
  - Over etched filling up empty tracks with metal shapes to meet Metal Density Rules
- Metal Slotting
  - Wide metal lines (Power Nets) expands significantly due to the high temperature during fabrication leads to destruction of the isolation and passivation layer that protect the wafer
  - To avoid it put slots or holes in these metal layers at regular intervals
  - Slotting also prevent the stress damage during wafer dicing and packaging

Formal Verification

- Formal Verification
  - Verify the two representations of circuit design exhibits same behavior
  - Checks the behavior of the Combinational Logics by checking the Compare Points
  - Targets implementation errors and not the design errors
  - Power checks: checks Power Switches/ Retention Cells/ Isolation Cells/ Level Shifters and all power connectivity
  - If any manual editing in the design then LEC has to be done at any point of time

- Formal Verification
  - Complete coverage
  - Effectively exhaustive simulation
  - Cover all possible sequences of inputs
  - Check all corner cases
  - No test vectors are needed

- Informal Verification (Simulation)
  - Incomplete coverage
  - Limited amount of simulation
  - Spot check a limited number of input sequences
  - Many corner cases not checked
  - Designer provides test vectors

Formal Verification

- Types of Formal Verification
  - Gate-level to Gate-level (Logical Equivalence Check after Routing)
    - To ensure that some netlist post-processing did not change the functionality of the circuit
  - RTL to Gate-level (after Synthesis)
    - To verify that the netlist correctly implements the original RTL code
  - RTL to RTL (before Synthesis)
    - To verify that two RTL descriptions are logically identical
- Logical Equivalence Check (LEC) will have two stages
  - Constrains setup stage
  - Logical Equivalence Check stage
- Tool will report equivalent/ non-equivalent/ abort/ not-checked
- Input Requirements
  - Netlists (.v)
  - Library (.lib and .lef)
  - Constraints (.sdc)
- Tools: Mentor Graphics FormalPro, Cadence Conformal, Synopsys Formality, Magma Quartz Formal

Parasitic Extraction

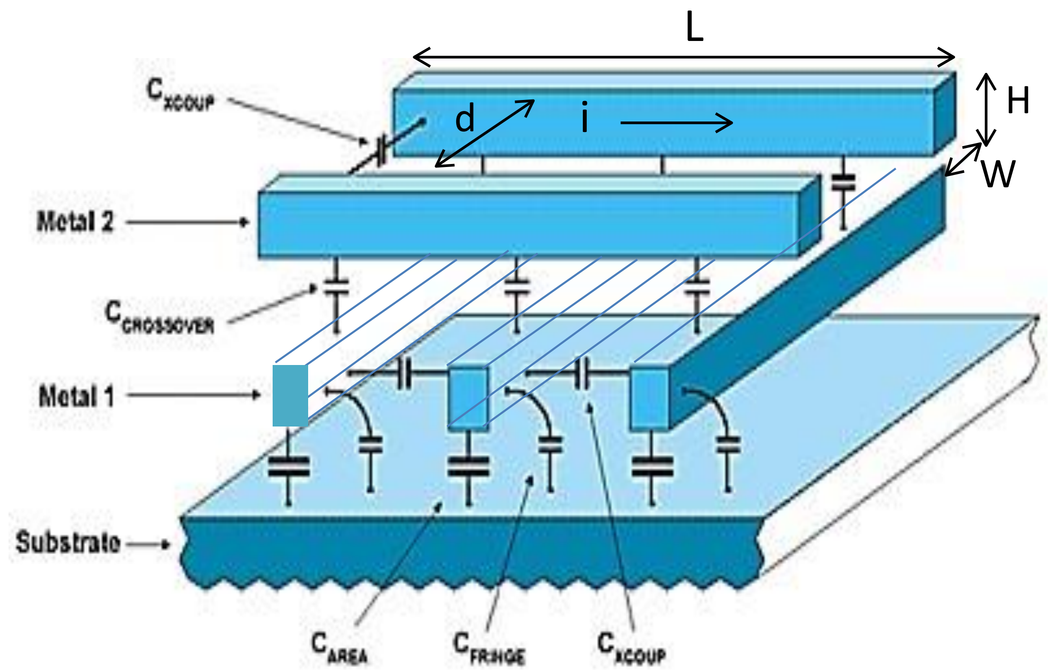
- Parasitic Extraction: Importance
  - Shrinking process geometries
  - New device structures
  - An increasing number of metal layers at each new process node
  - Much more closer nets at each new process node
  - Increasing wire aspect ratio of height to width
  - Increasing operating frequency
- Parasitic Capacitance can be reduced by using higher metals, provide spacing, shielding, Avoid parallel routing
- At higher clock frequencies, RC interconnect modeling is no longer adequate and inductance must be included in interconnect modeling
- Reluctance (Inductance) effect becomes more and more prominent as the resistance (both device and interconnect) decreases and the operating frequency increases

Parasitic Extraction

- Capacitance

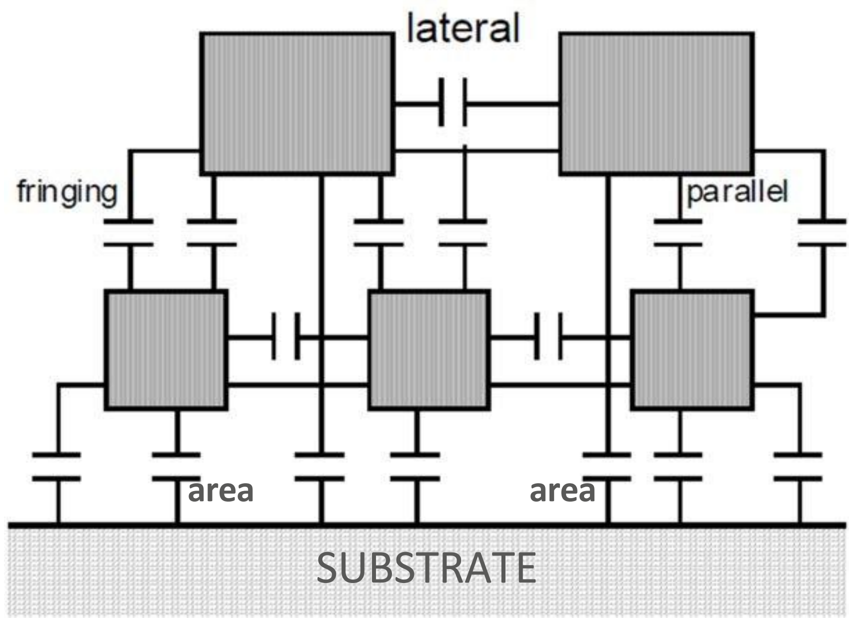
  $C = \varepsilon_o W H/d$

  - Transistors
    - Depends on area of transistor gate, physical of materials, thickness of insulator, diffusion to substrate
  - Poly to Substrate
    - Parallel plate and fringing
  - Capacitance between conductors
    - Coupling Capacitance
    - Area Capacitance
    - Fringing Capacitance
    - Crossover Capacitance



Parasitic Extraction

- Coupling Capacitance/ Lateral Capacitance
  - The capacitance between nets on the same Metal layer
  - Dominant over interlayer capacitances with every new process technology
- Fringing Capacitance
  - Capacitance between nets of different Metal layers and other layers due to Sidewall Capacitance
- Parallel/Crossover Capacitance
  - Capacitance between nets of 2 different Metal layers



- Area Capacitance
  - Capacitance between Metal layers and Substrate
- In modern processes, the width of interconnect wires at lower levels of metal is so small that the Fringing Capacitance of the wire is larger than the Area Capacitance

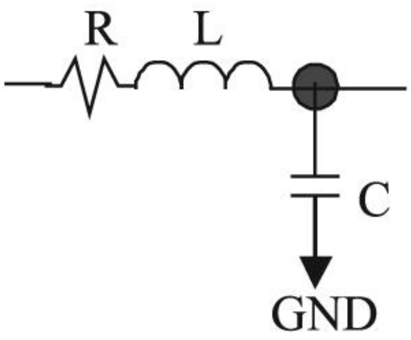
Parasitic Extraction

- Resistance

  R = ρ L/H W

  - Wire Resistivity
  - Complex 3D geometry around Vias

- Inductance
  - Self Inductance; $V = L\frac{di}{dt}$
  - Mutual Inductance, $M = K\sqrt{L_1L_2}$
  - At high frequency Skin effect possibility



- Models used for Parasitic Extraction
  - Lumped-C, Lumped-RC, Lumped-RLC
  - Pi segment
  - Pin-to-pin delays are modeled by RC delays

Timing Analysis

- Static Timing Analysis: Methodical analysis of a digital circuit to determine if the timing constraints imposed are met and to check the design is working properly
- Static Timing Analysis Flow
  - Read the inputs required
  - Setting up Constraints: IO Delay Constraints, DRVs, Timing Exceptions (False/ Multi-Cycle paths), Recovery and Removal, Minimum Pulse Width
  - Construct Timing Graph: Partition Clock Domain, Ideal/ Propagated Clock, Case Analysis
  - Propagation
  - Timing Report: End points with violations/ Paths enumeration
- Input Requirement
  - Routed Netlist (.v)
  - Libraries (.lib only)
  - Constraints (.sdc)
  - Delay Format (.sdf)
  - Parasitic Values (.spef)
- Tools: Synopsys PrimeTime, Cadence ETS, Cadence Tempus

Power Analysis & IR Drop Analysis

- Power Analysis
  - Static/ Leakage Power Analysis
  - Dynamic Power Analysis
- IR Drop Analysis
  - Static IR Drop Analysis
  - Dynamic IR Drop Analysis
- Tools for Power and IR Drop Analysis
  - Synopsys Prime Power
  - Cadence EPS and Voltus
  - Apache Redhawk
- Tape-out
  - Final GDSII (Graphical Data Stream Information Interchange) or CIF (Caltech Intermediate Format) to Foundry
  - GDS contains Physical Layout information

Thank You